

BFS Traversal

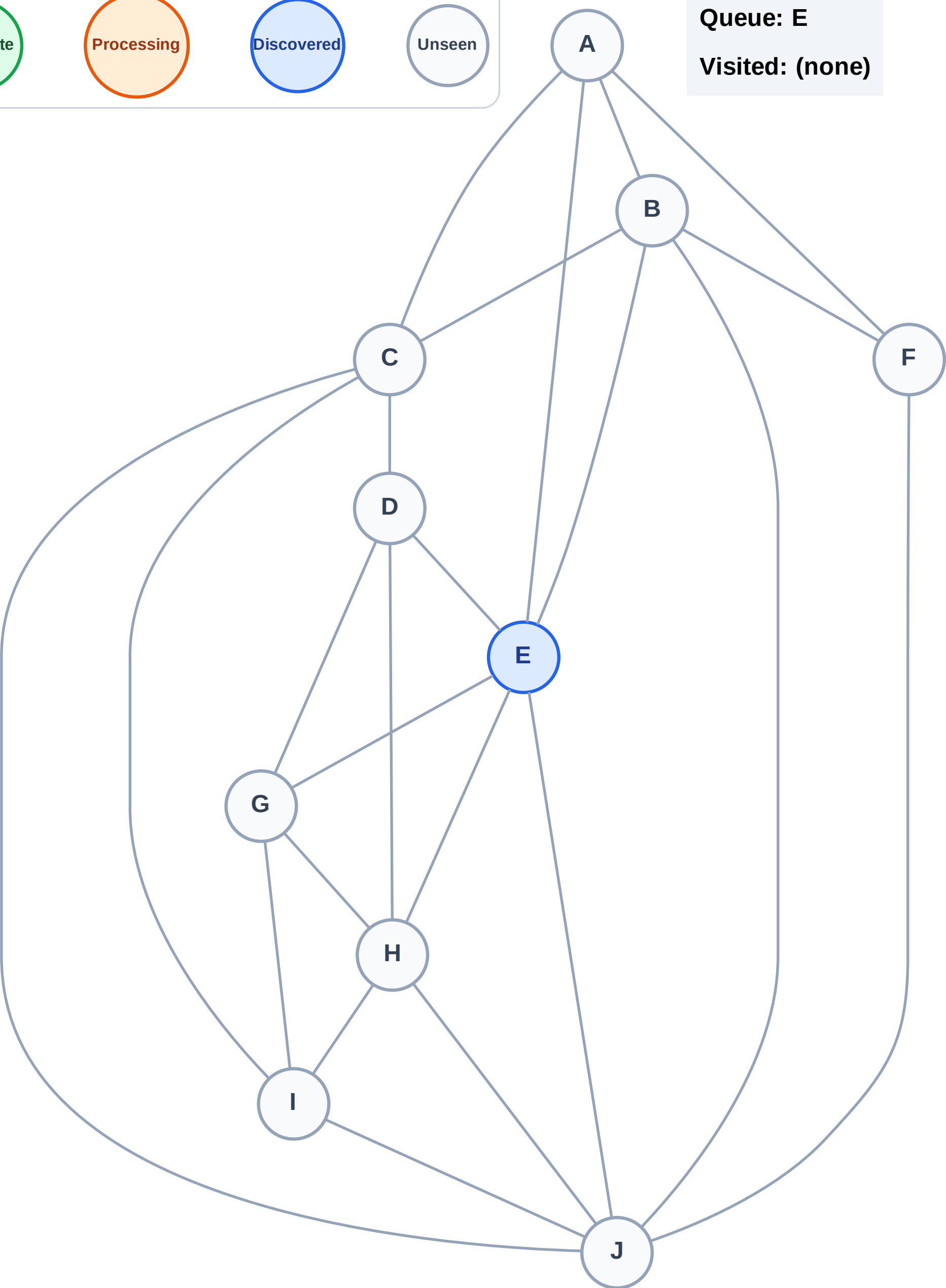
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

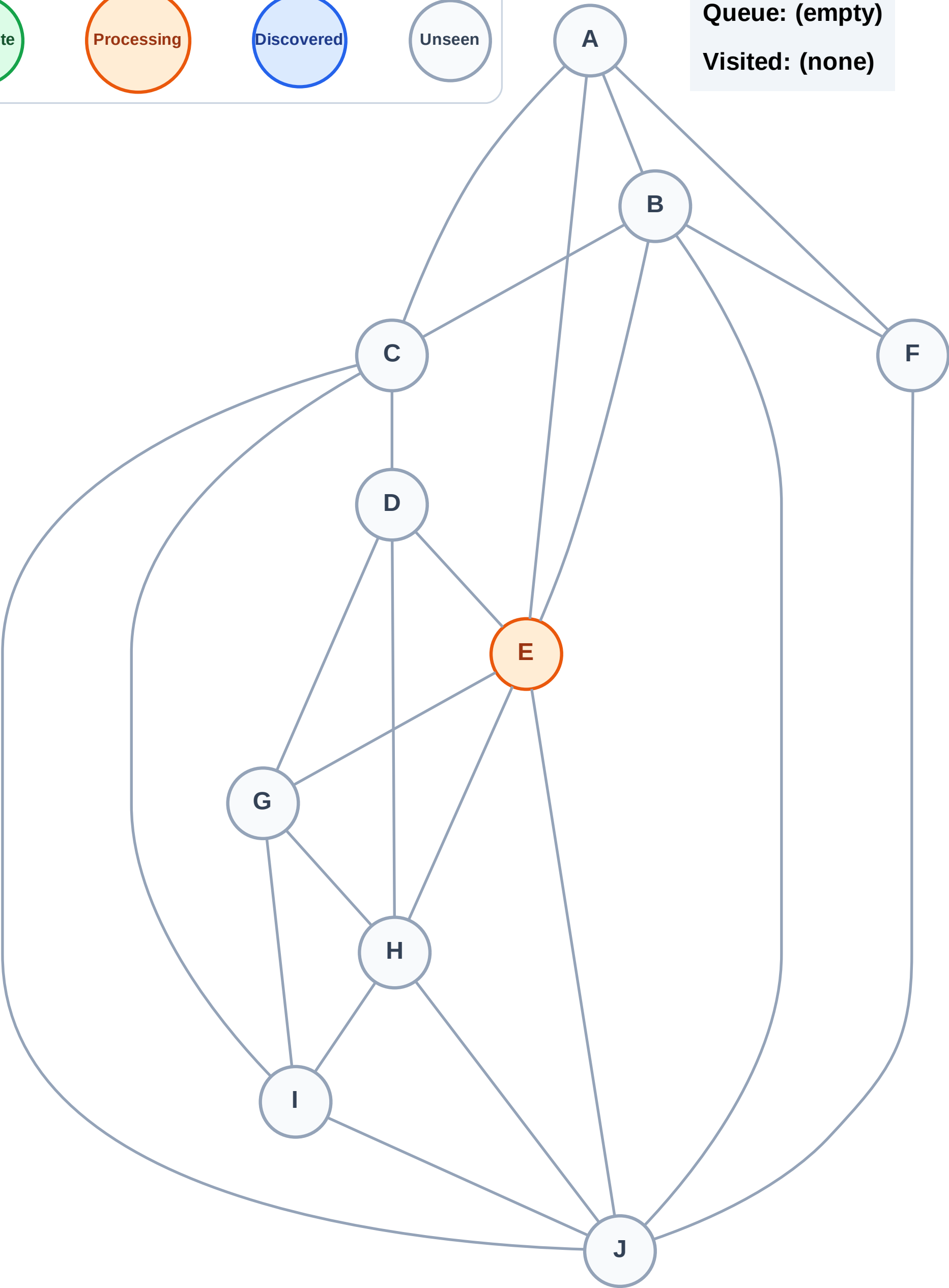
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover A, B, D, G, H, J from E

Legend

Complete

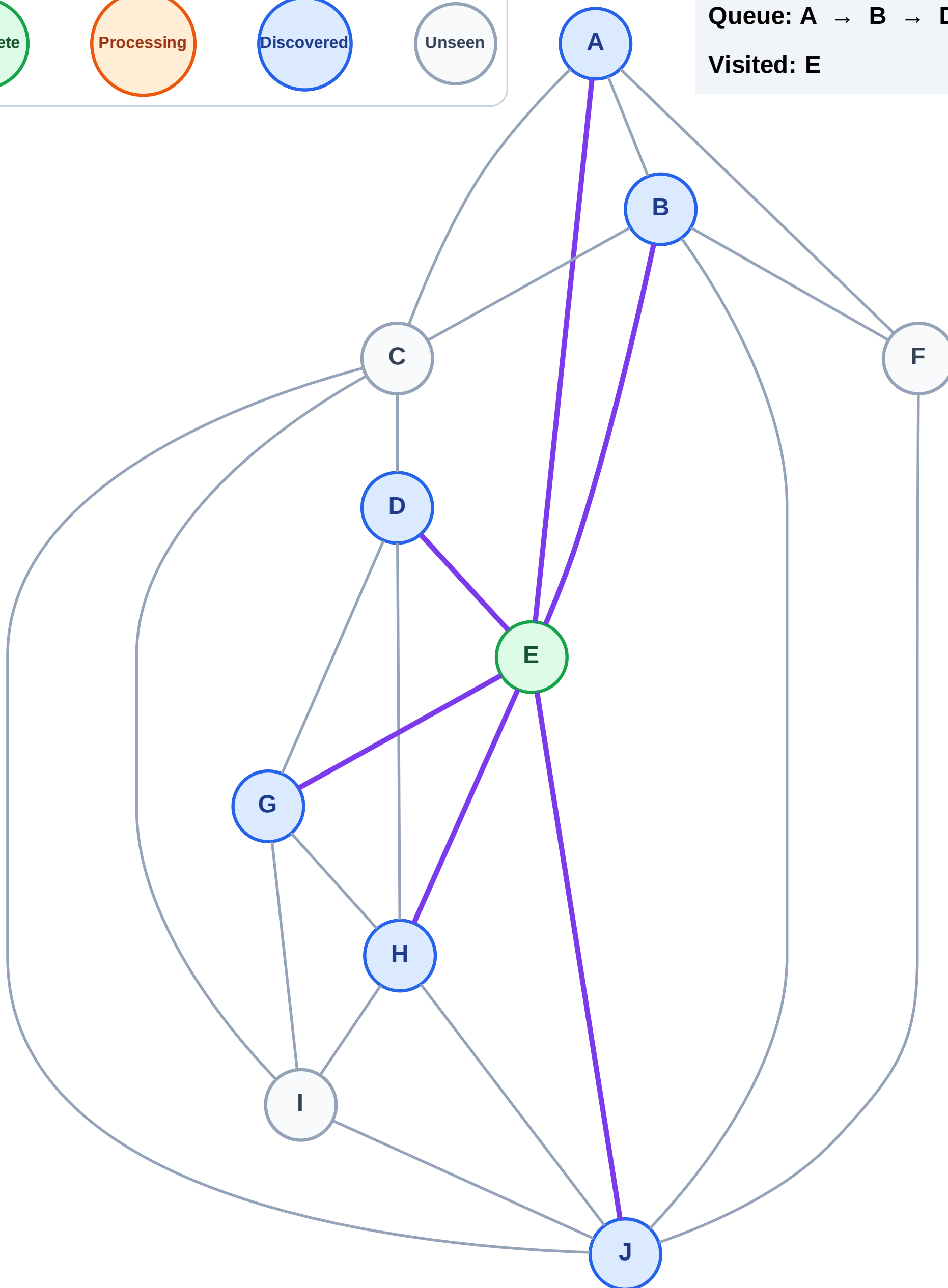
Processing

Discovered

Unseen

Queue: A → B → D → G → H → J

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

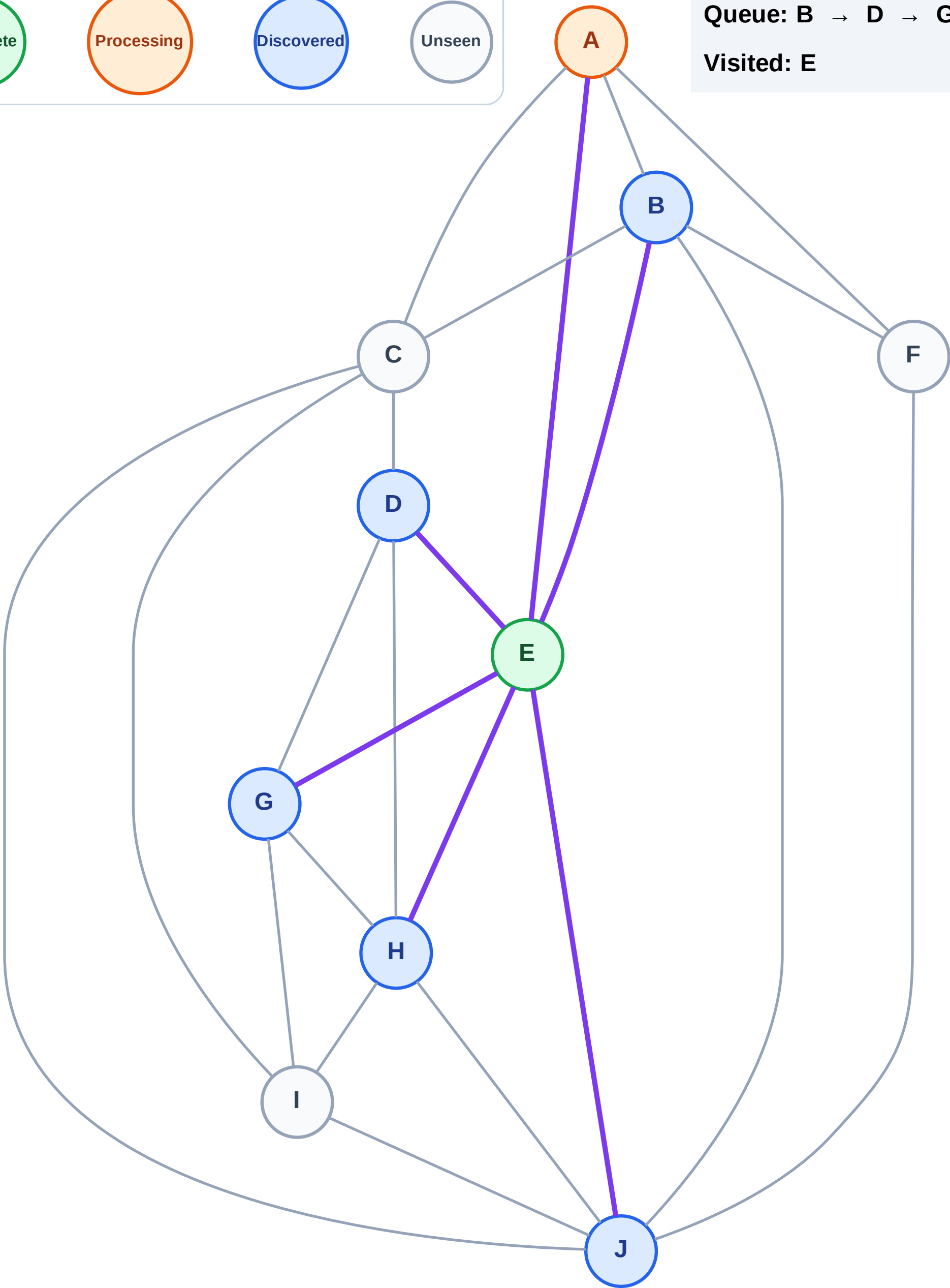
Processing

Discovered

Unseen

Queue: B → D → G → H → J

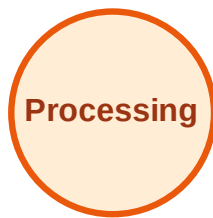
Visited: E



BFS Traversal

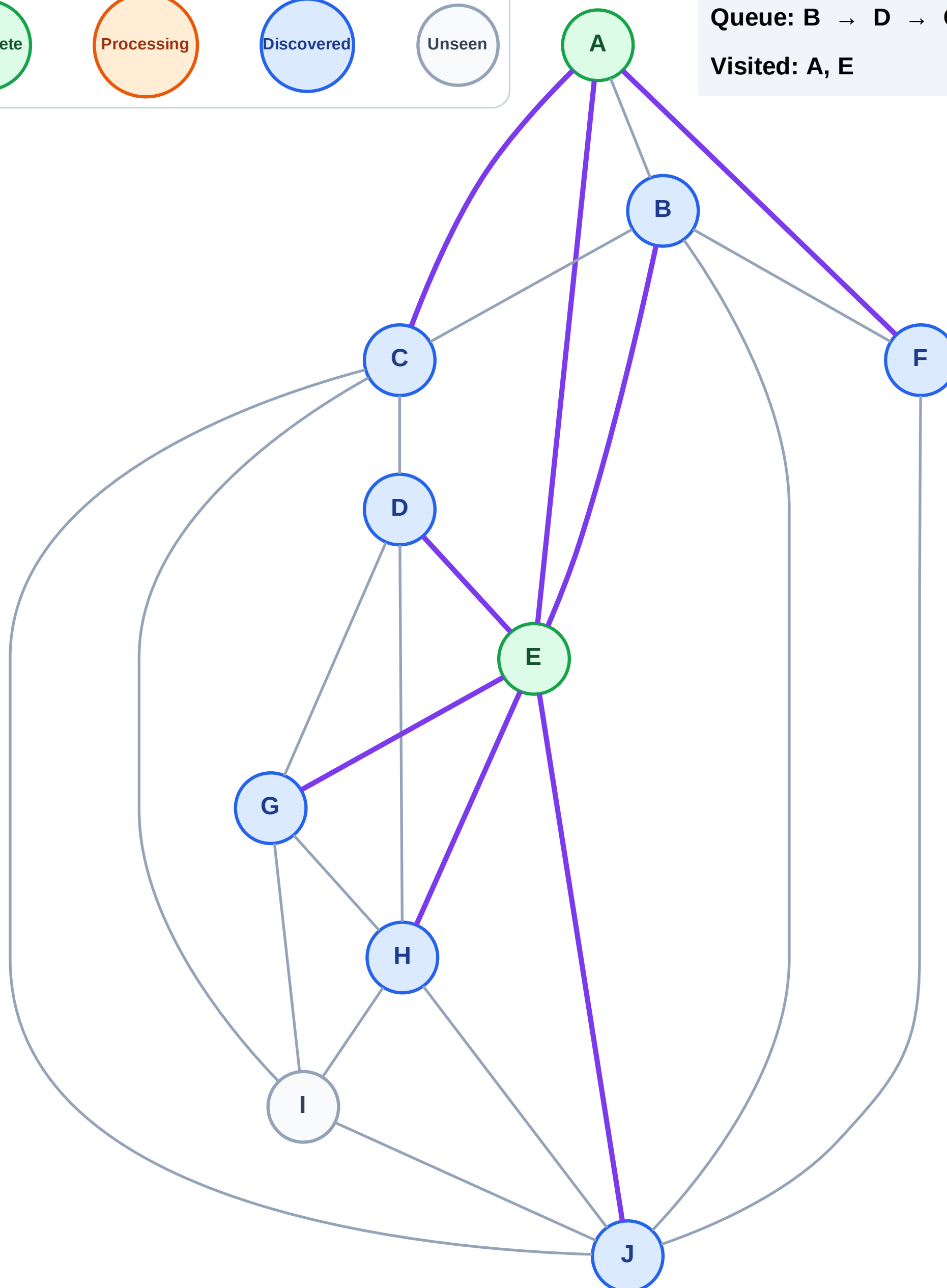
Step 5: Discover C, F from A

Legend



Queue: B → D → G → H → J → C → F

Visited: A, E



BFS Traversal

Step 6: Process node B

Legend

Complete

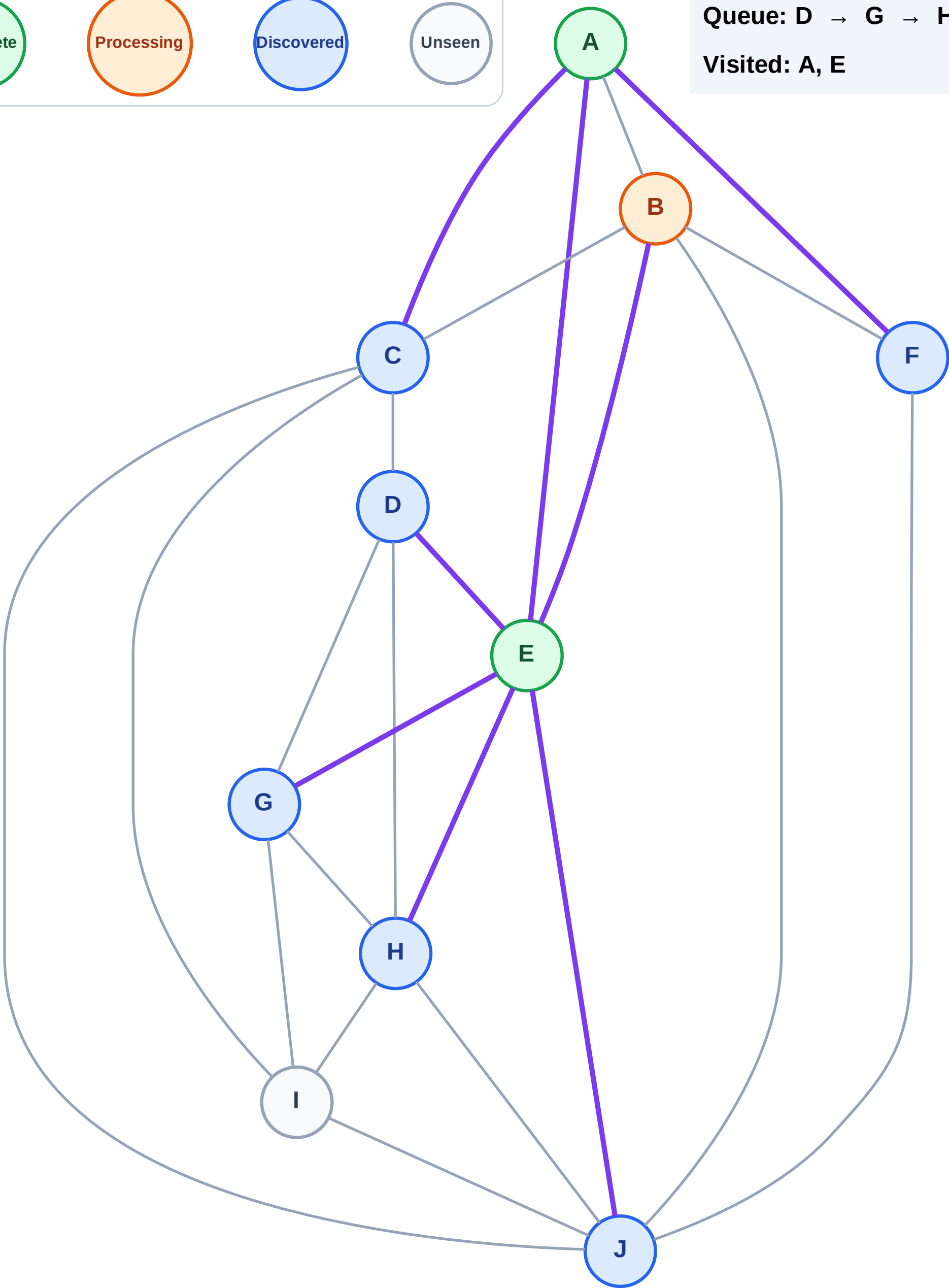
Processing

Discovered

Unseen

Queue: D → G → H → J → C → F

Visited: A, E



BFS Traversal

Step 7: No new neighbors from B

Legend

Complete

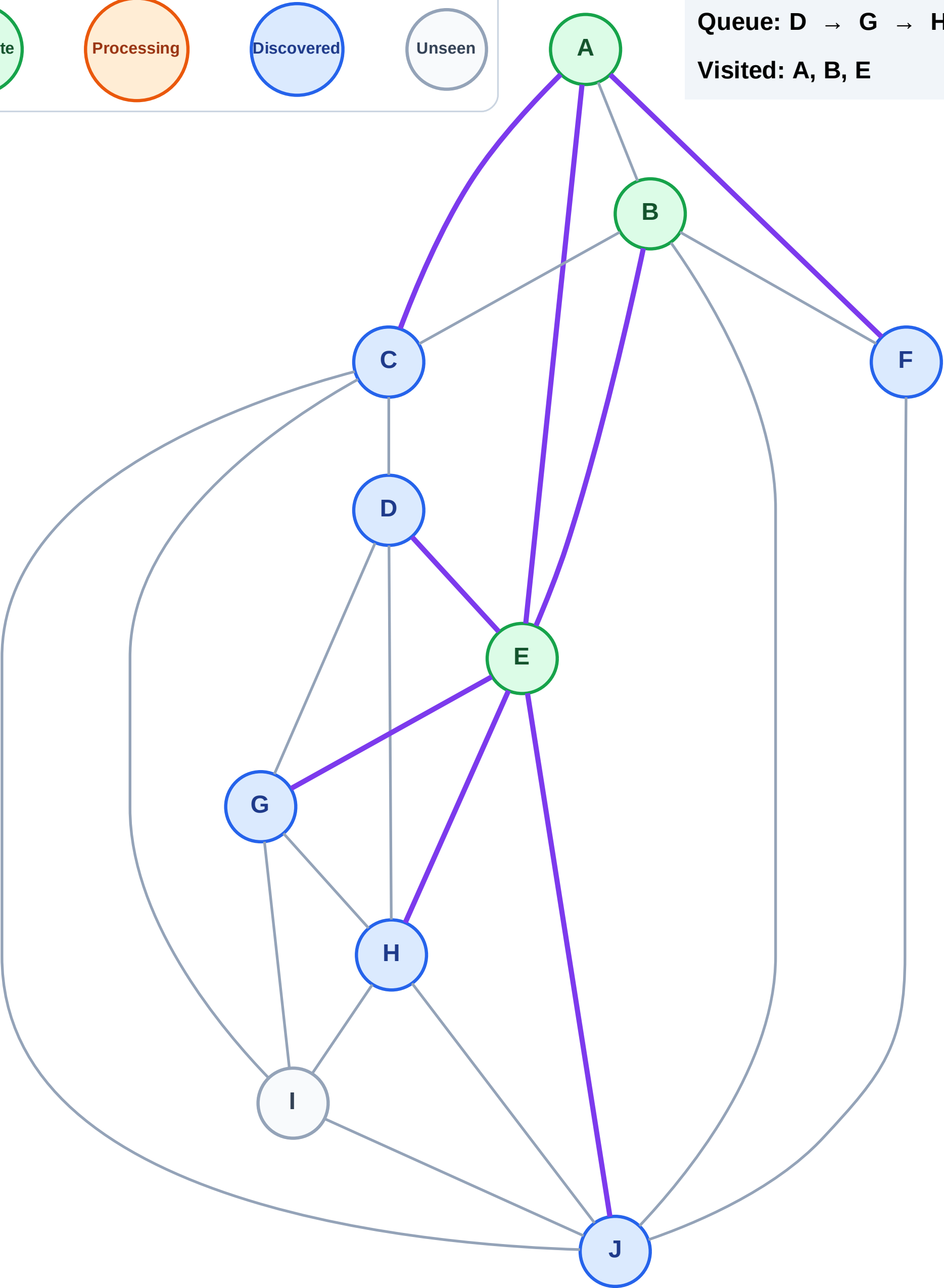
Processing

Discovered

Unseen

Queue: D → G → H → J → C → F

Visited: A, B, E



BFS Traversal

Step 8: Process node D

Legend

Complete

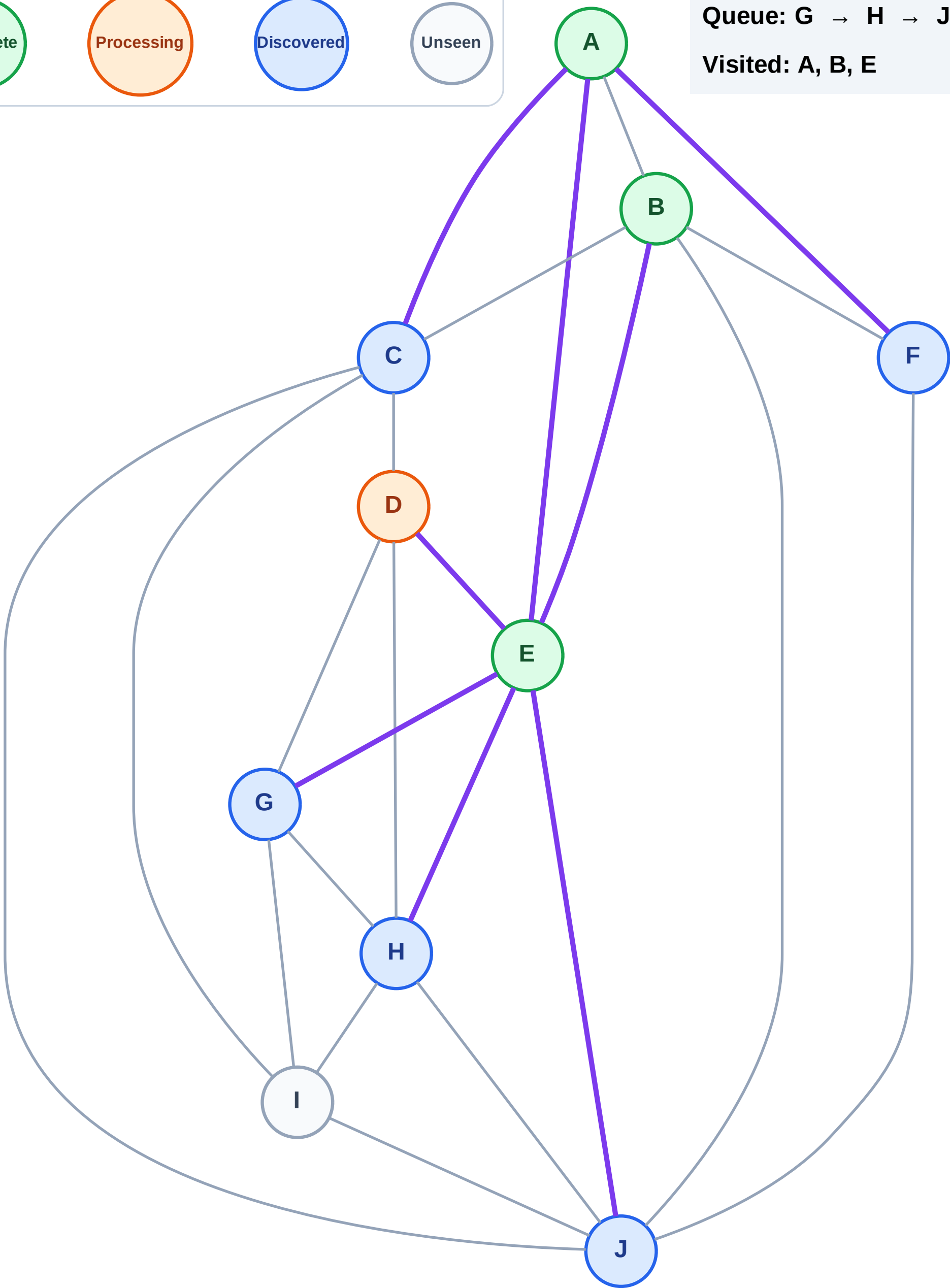
Processing

Discovered

Unseen

Queue: G → H → J → C → F

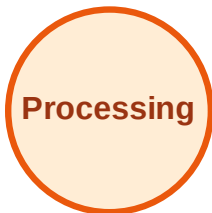
Visited: A, B, E



BFS Traversal

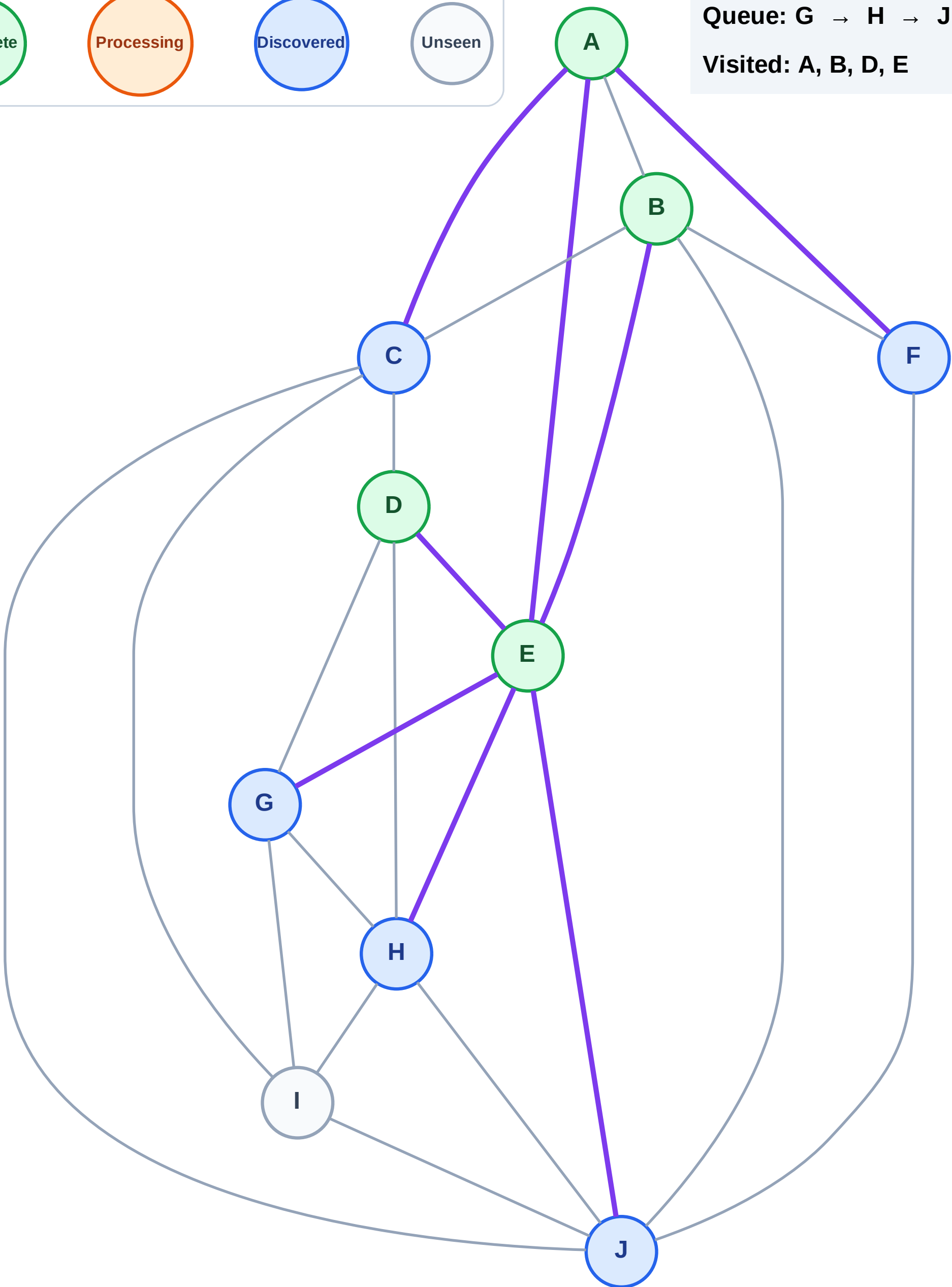
Step 9: No new neighbors from D

Legend



Queue: G → H → J → C → F

Visited: A, B, D, E



BFS Traversal

Step 10: Process node G

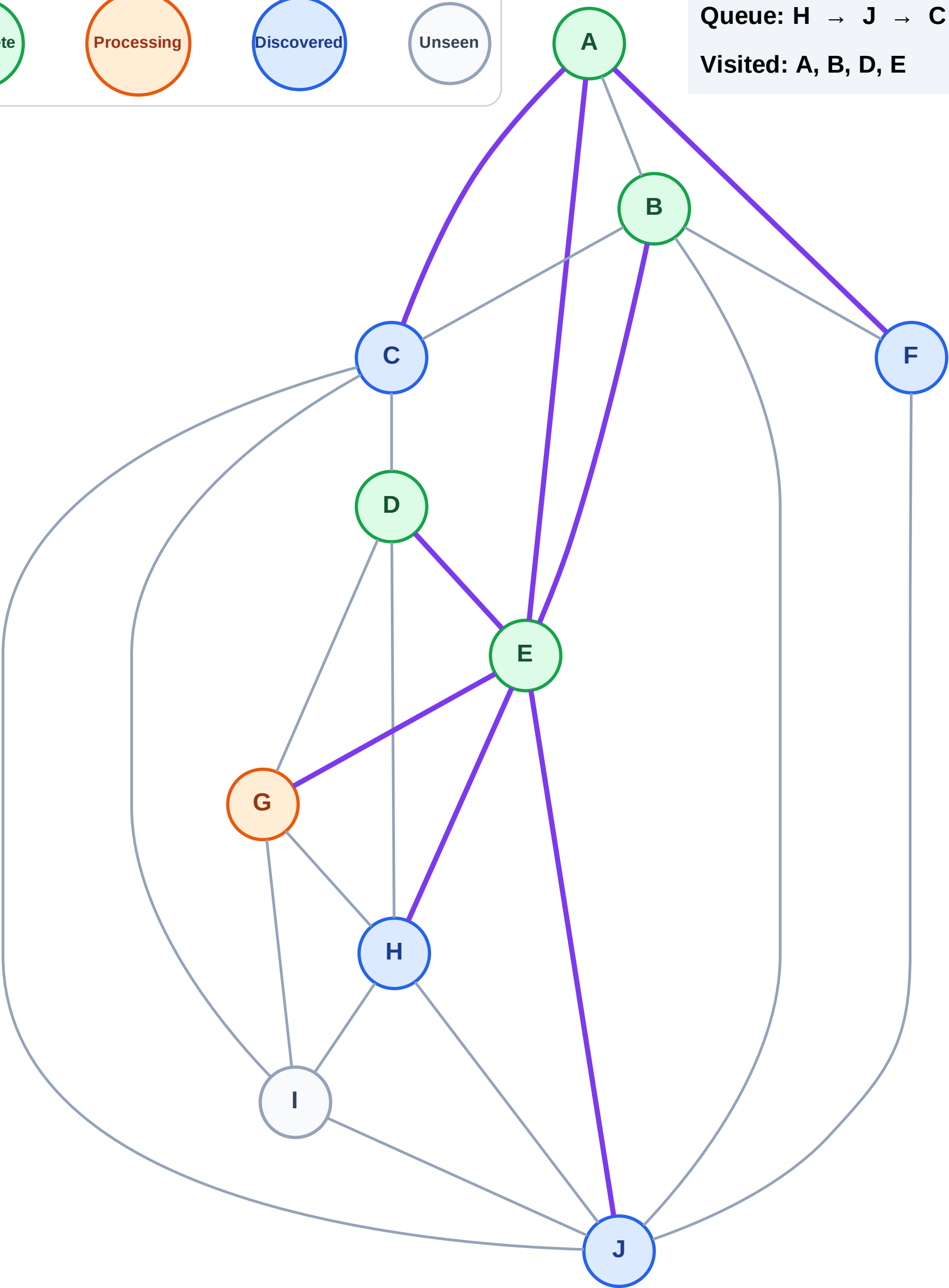
Complete

Processing

Discovered

Unseen

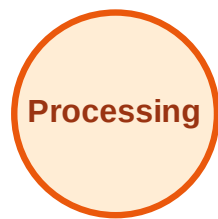
Queue: H → J → C → F
Visited: A, B, D, E



BFS Traversal

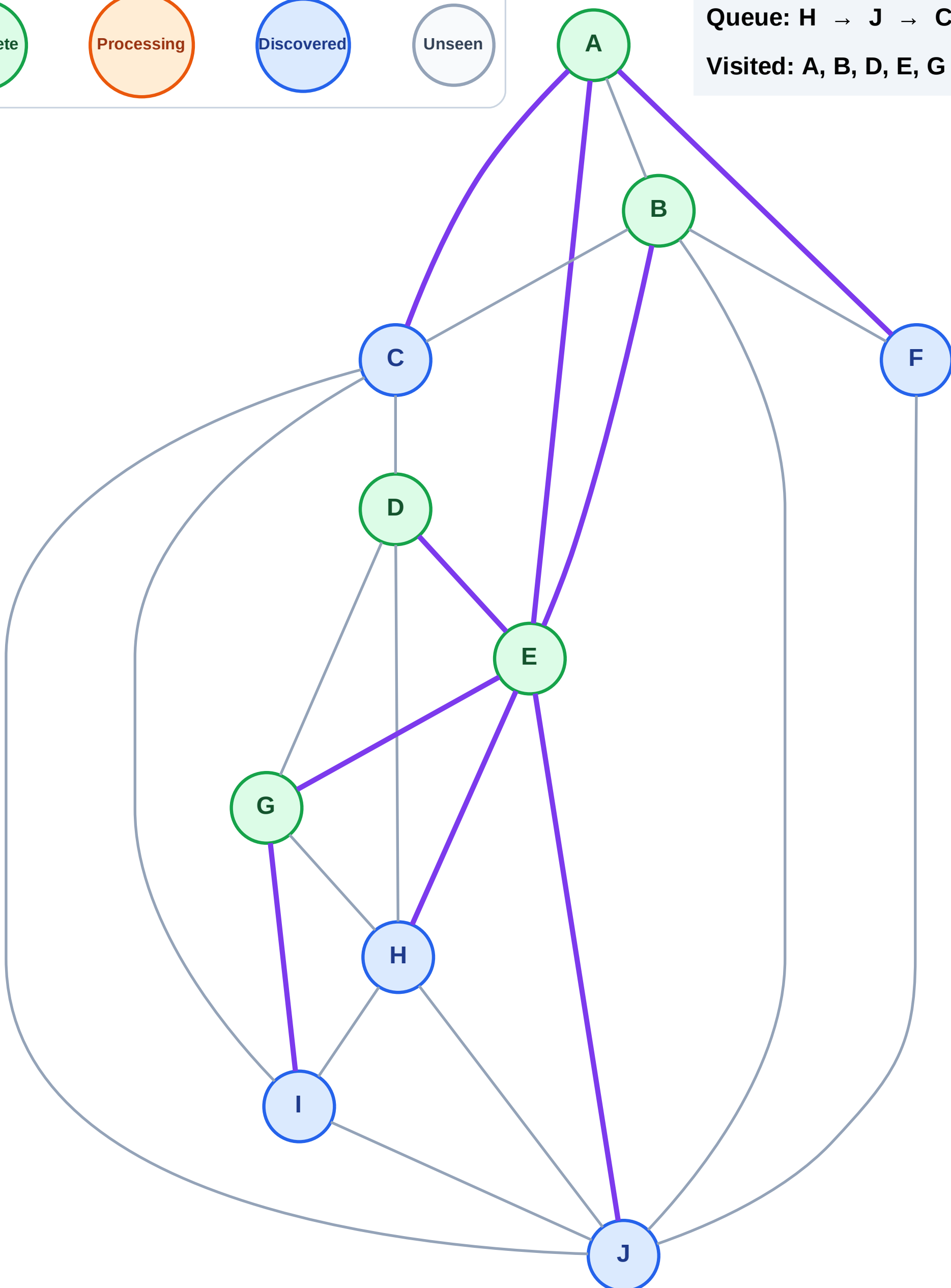
Step 11: Discover I from G

Legend



Queue: H → J → C → F → I

Visited: A, B, D, E, G



BFS Traversal

Step 12: Process node H

Legend

Complete

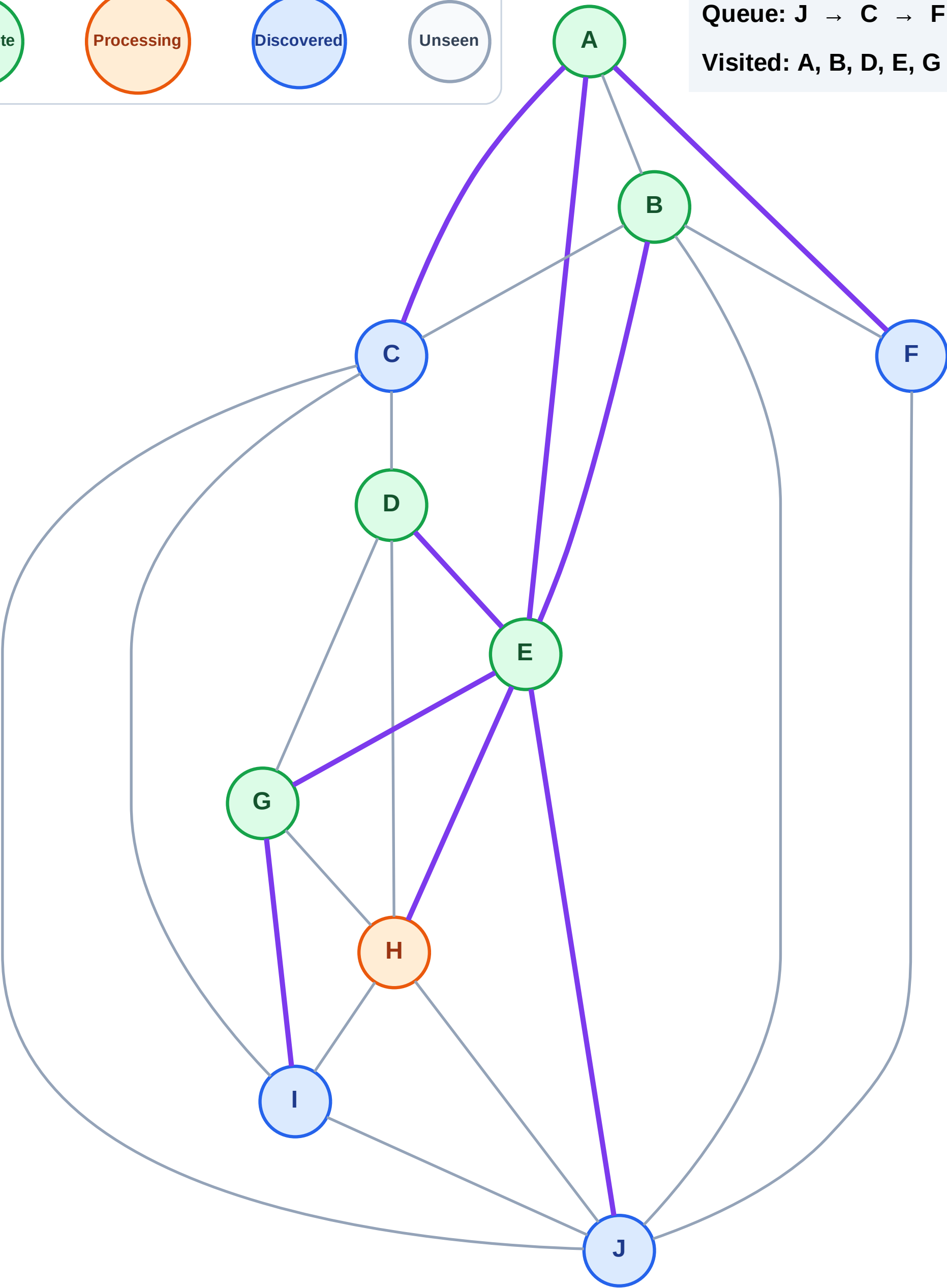
Processing

Discovered

Unseen

Queue: J → C → F → I

Visited: A, B, D, E, G



BFS Traversal

Step 13: No new neighbors from H

Complete

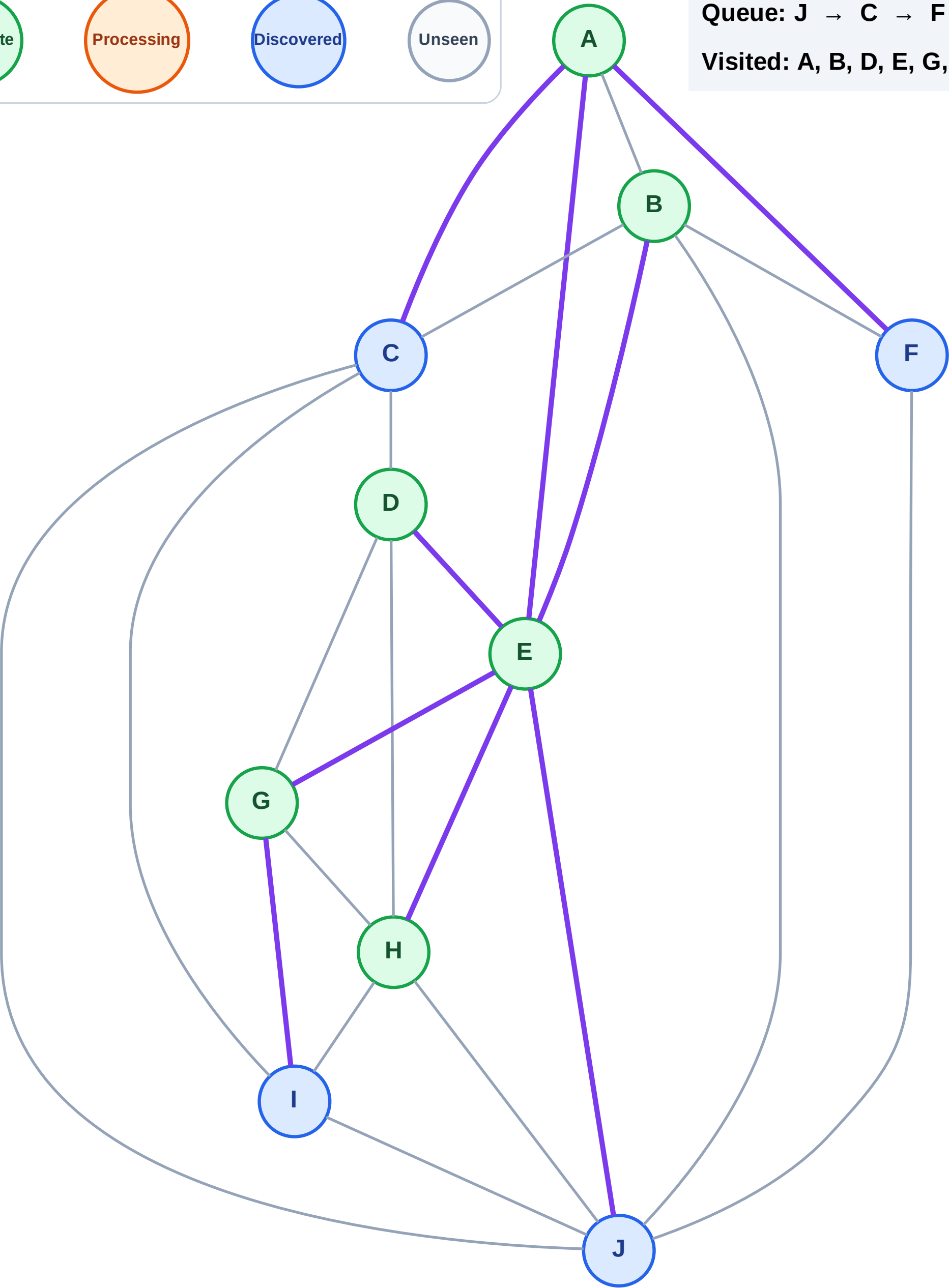
Processing

Discovered

Unseen

Queue: J → C → F → I

Visited: A, B, D, E, G, H



BFS Traversal

Step 14: Process node J

Legend

Complete

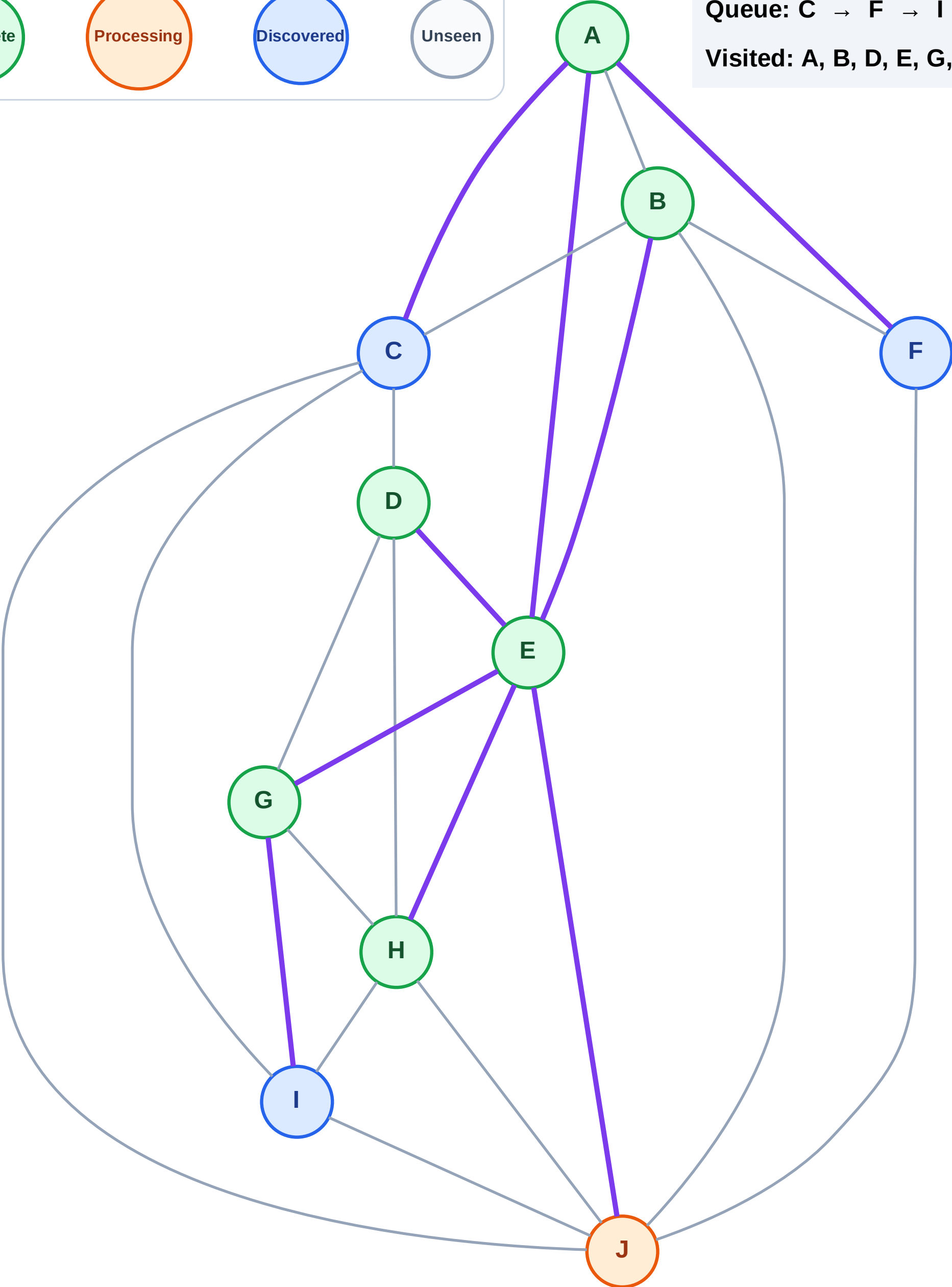
Processing

Discovered

Unseen

Queue: C → F → I

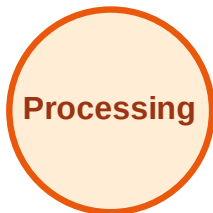
Visited: A, B, D, E, G, H



BFS Traversal

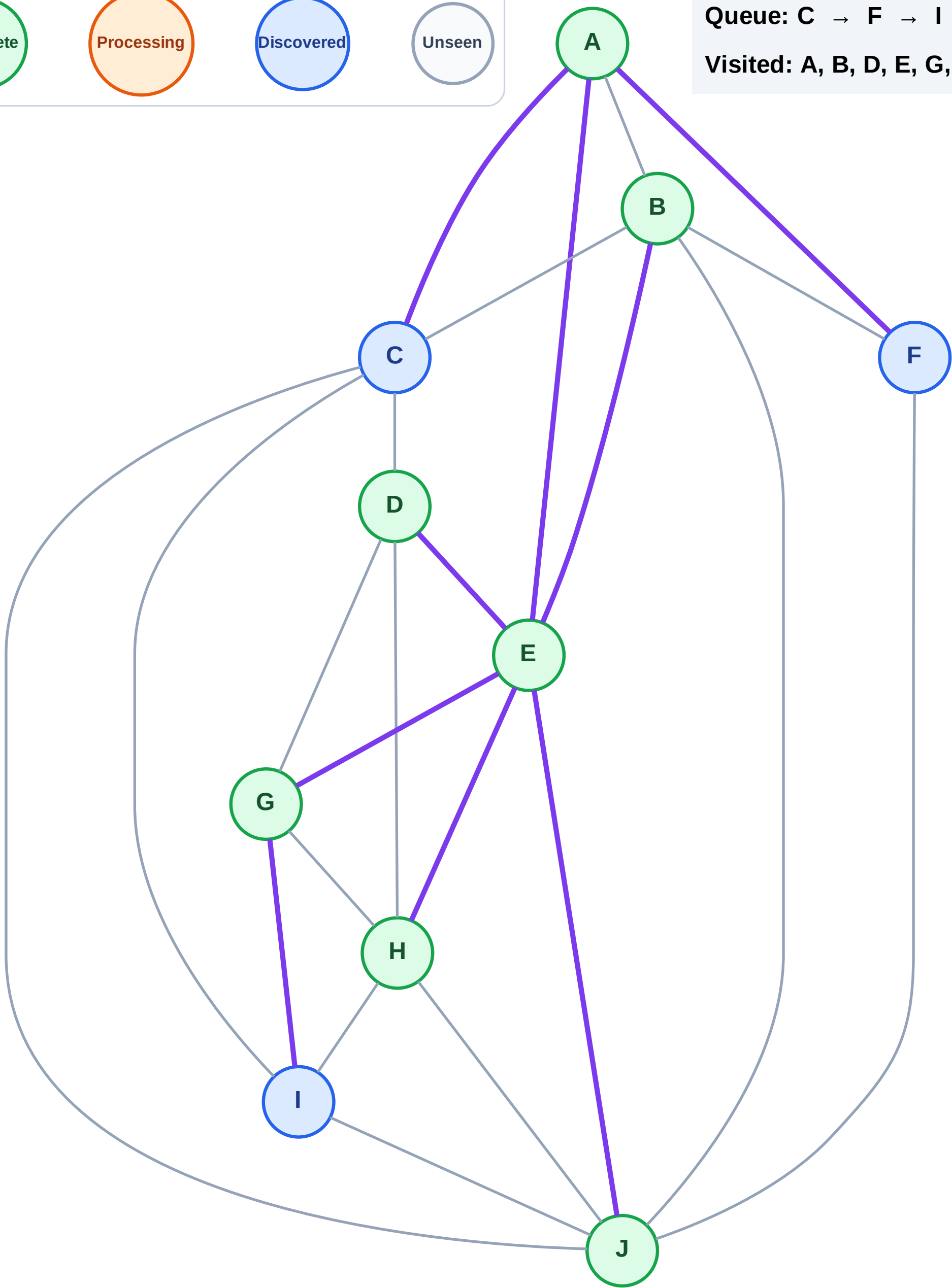
Step 15: No new neighbors from J

Legend



Queue: C → F → I

Visited: A, B, D, E, G, H, J



BFS Traversal

Step 16: Process node C

Legend

Complete

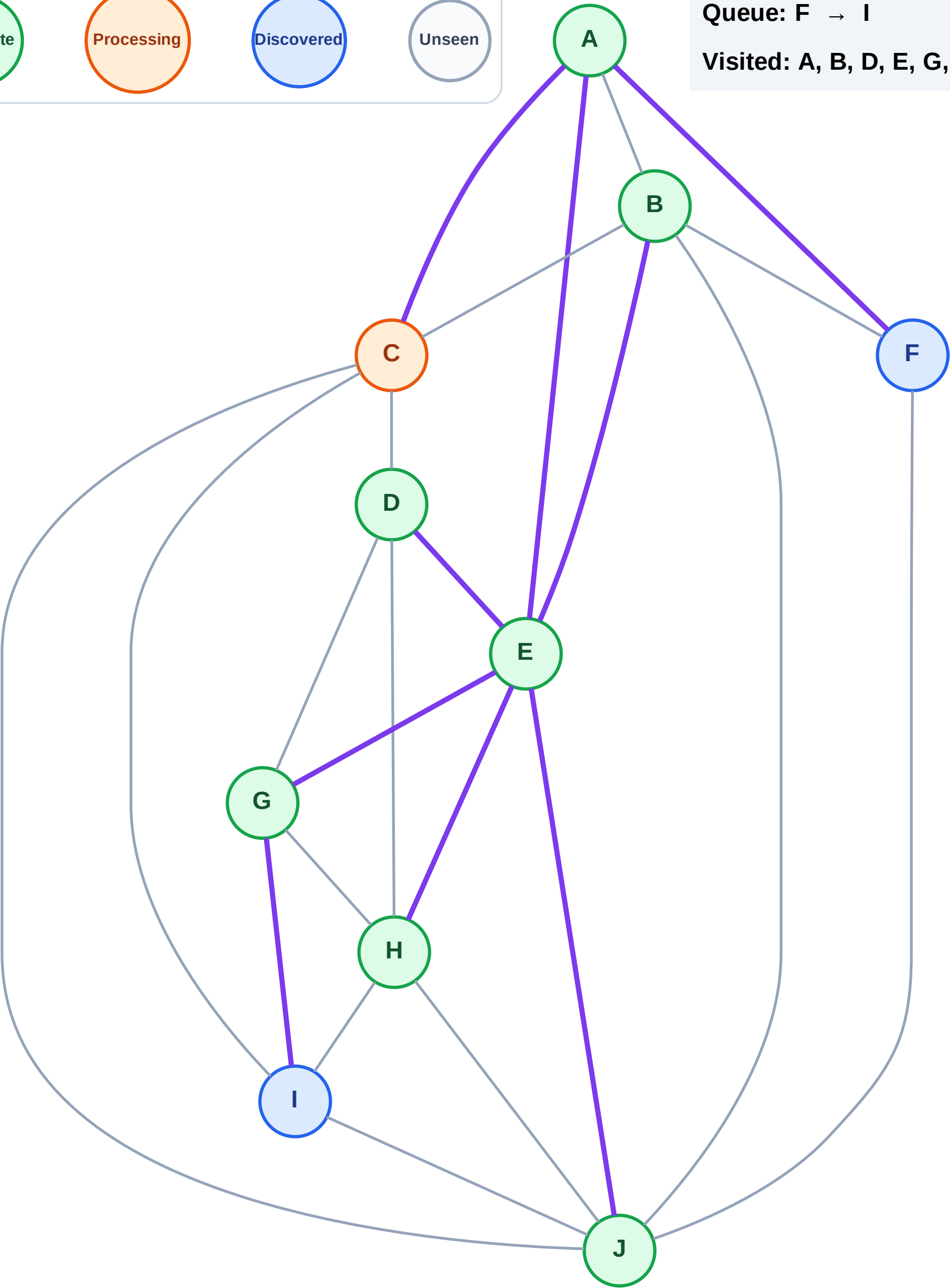
Processing

Discovered

Unseen

Queue: F → I

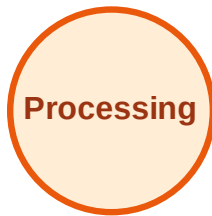
Visited: A, B, D, E, G, H, J



BFS Traversal

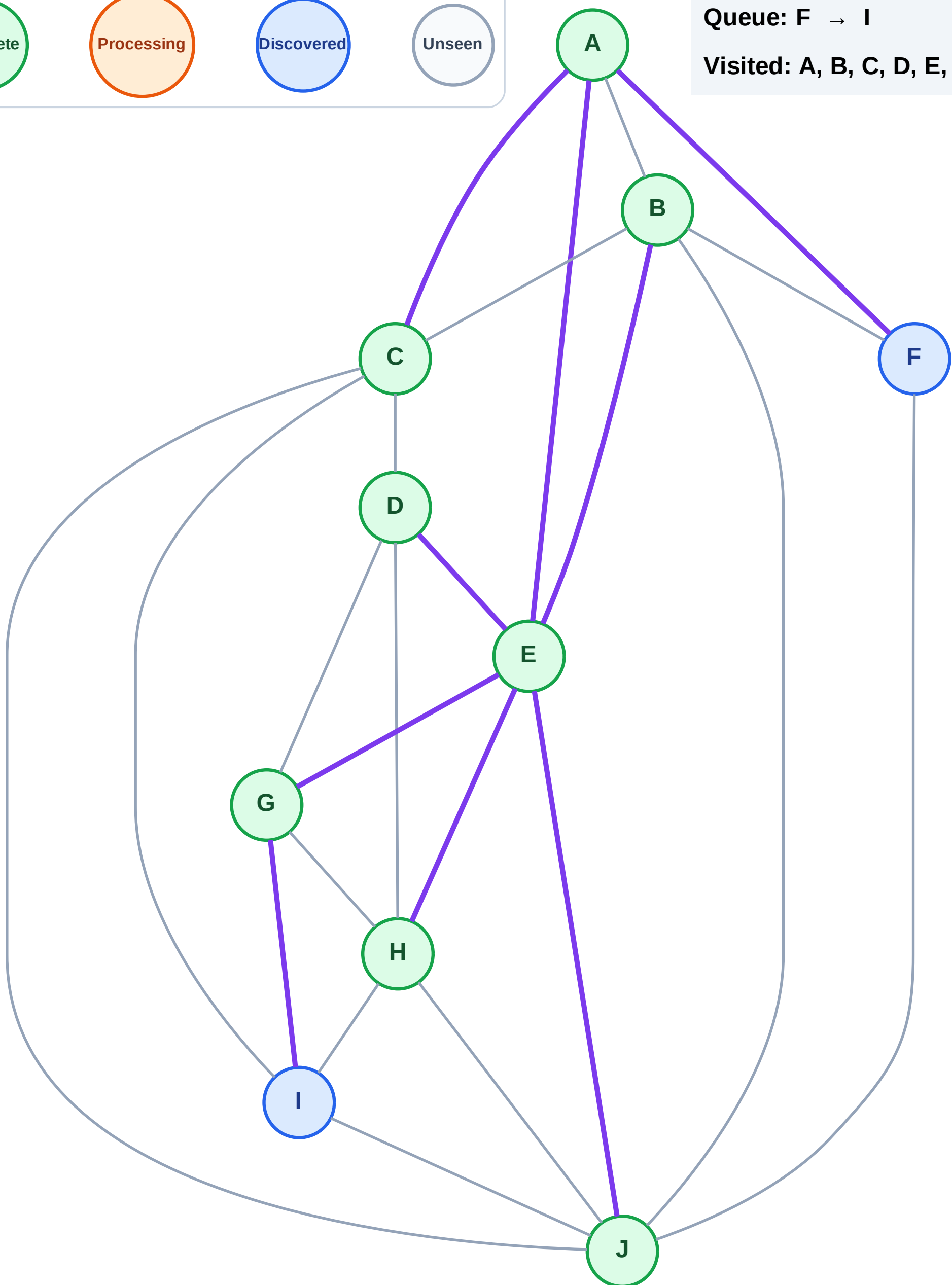
Step 17: No new neighbors from C

Legend



Queue: F → I

Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 18: Process node F

Legend

Complete

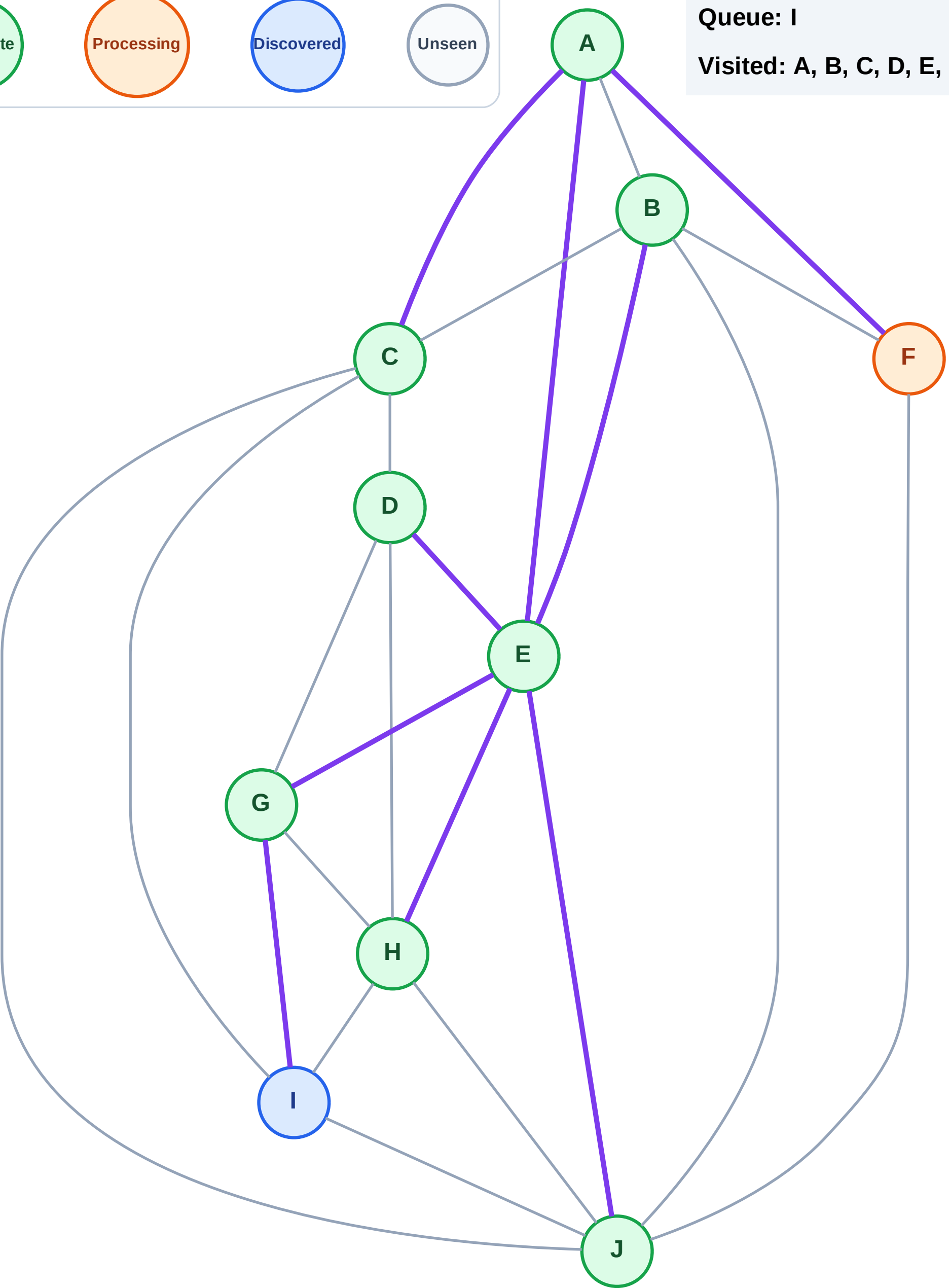
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, G, H, J



BFS Traversal

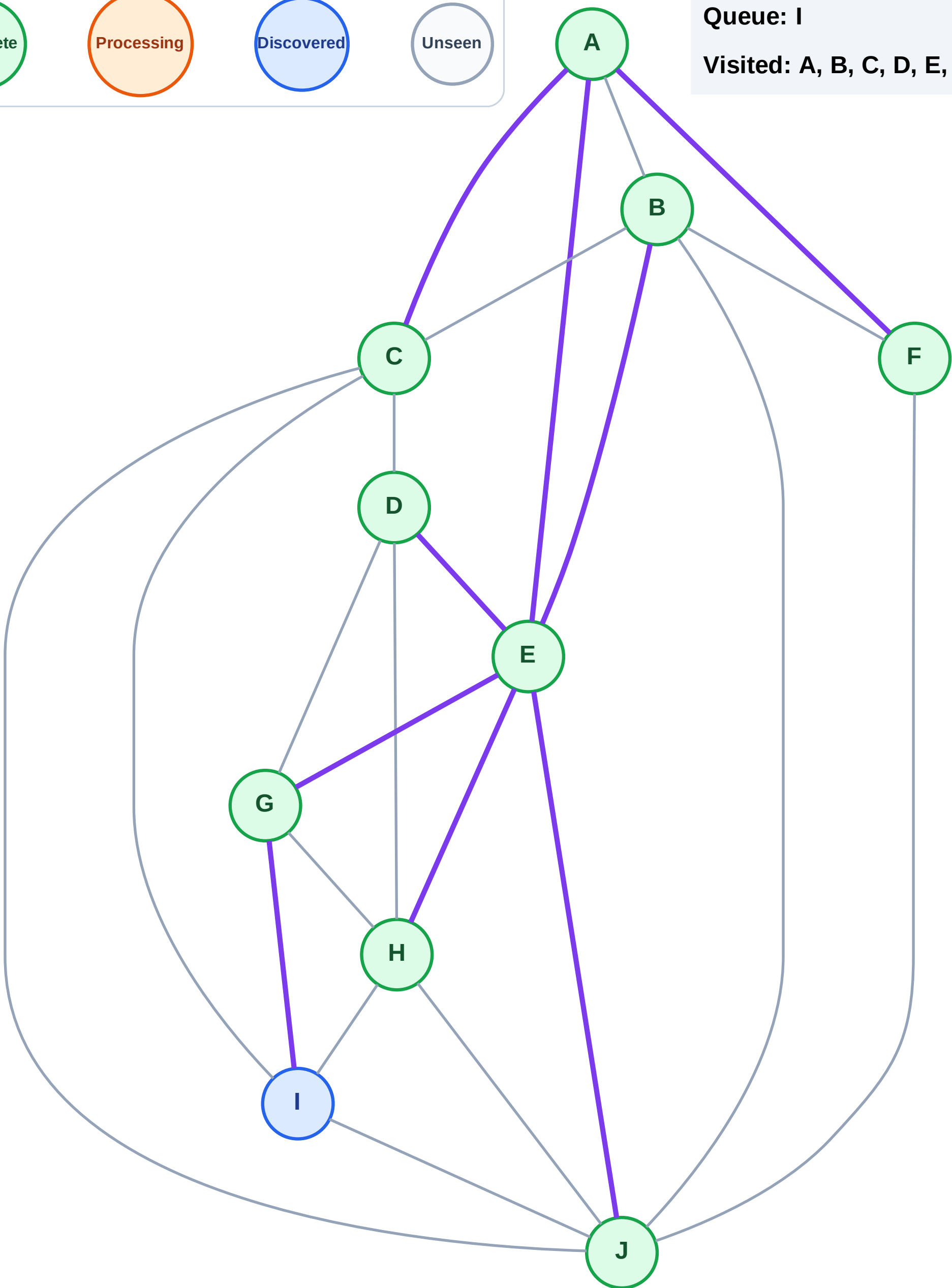
Step 19: No new neighbors from F

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

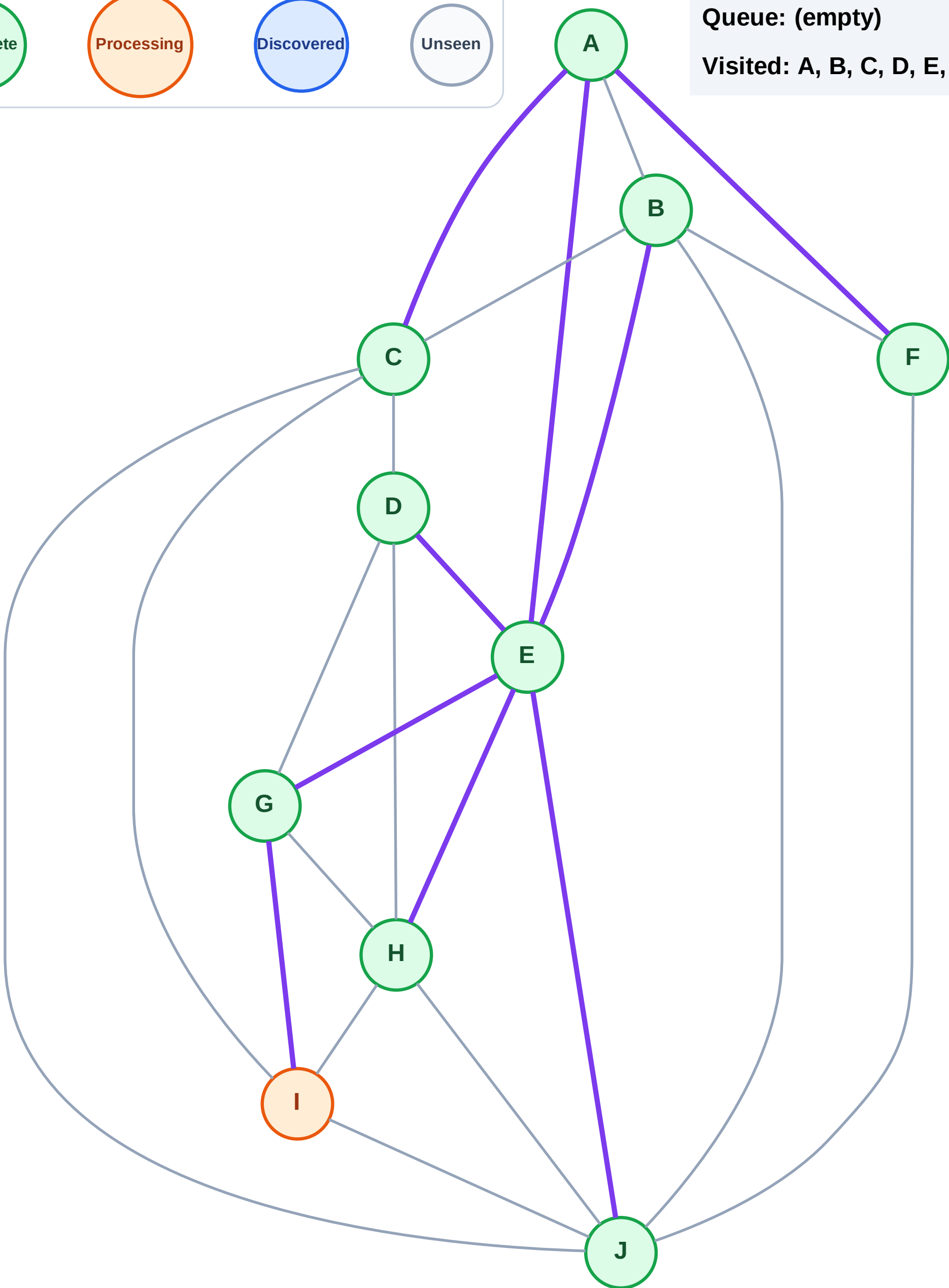
Step 20: Process node I

Legend



Queue: (empty)

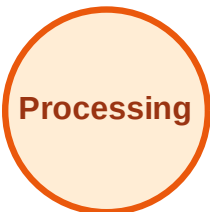
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

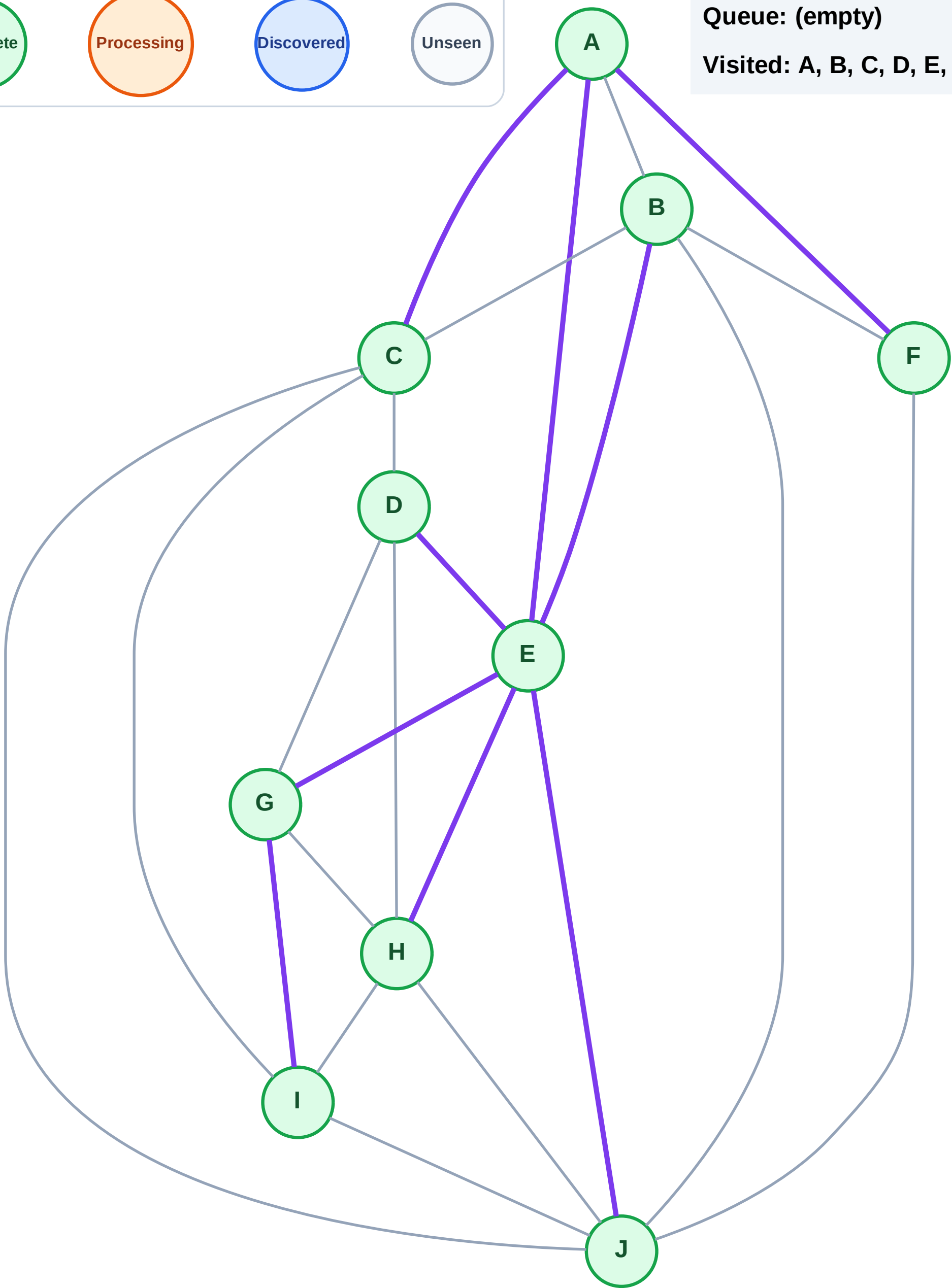
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

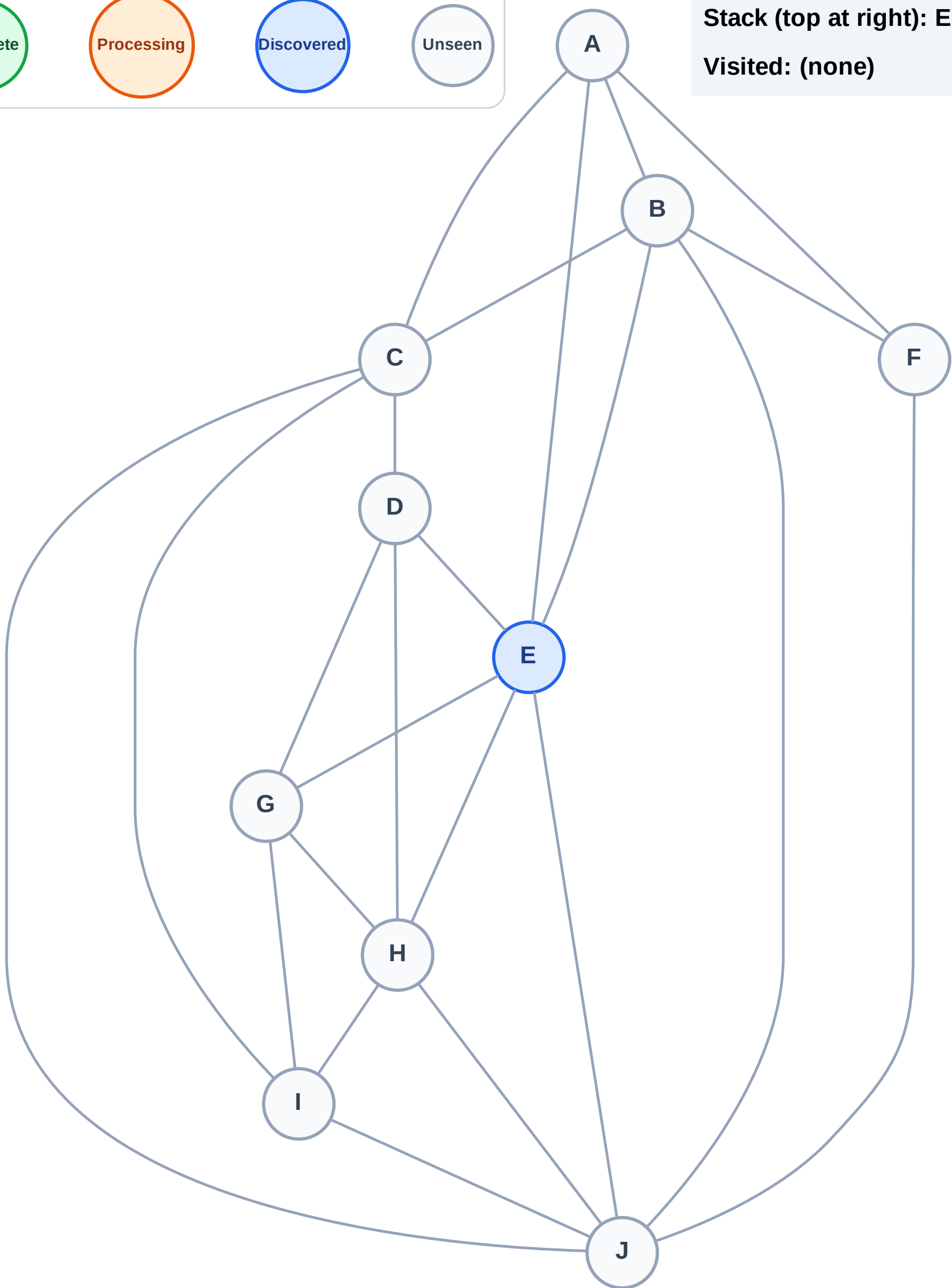
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

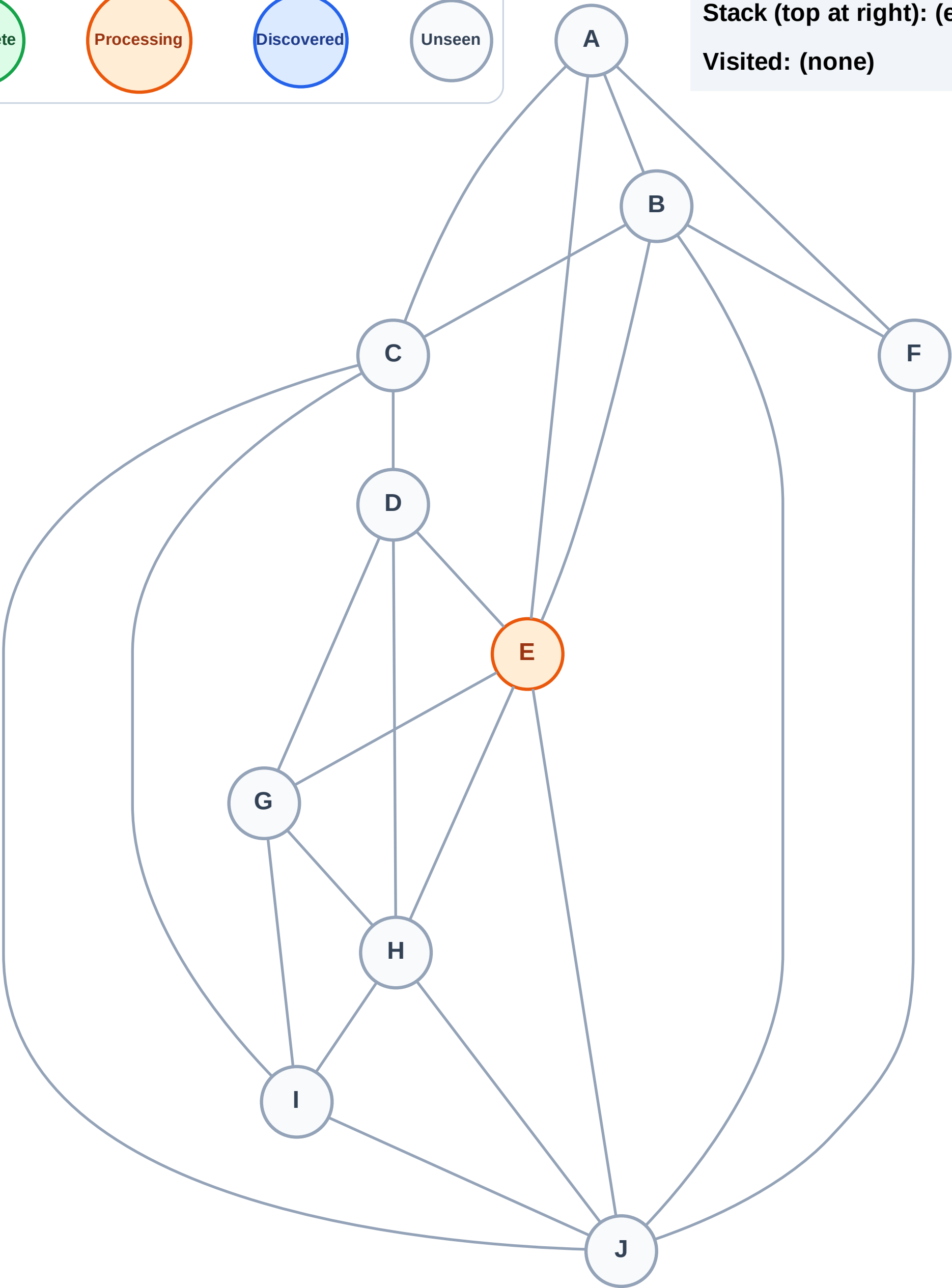
Complete

Processing

Discovered

Unseen

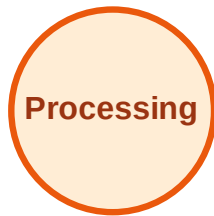
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

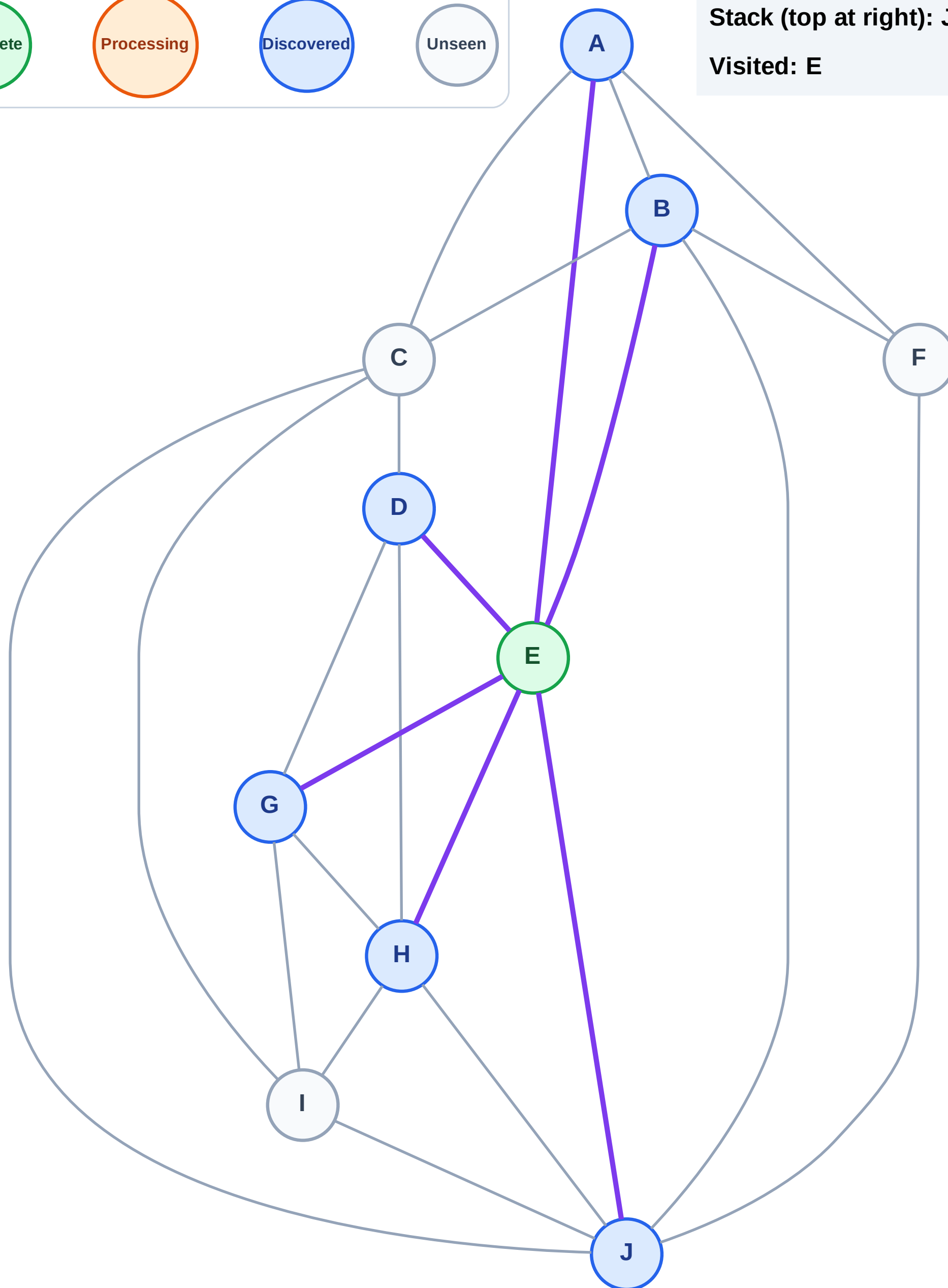
Step 3: Discover J, H, G, D, B, A from E

Legend



Stack (top at right): J | H | G | D | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

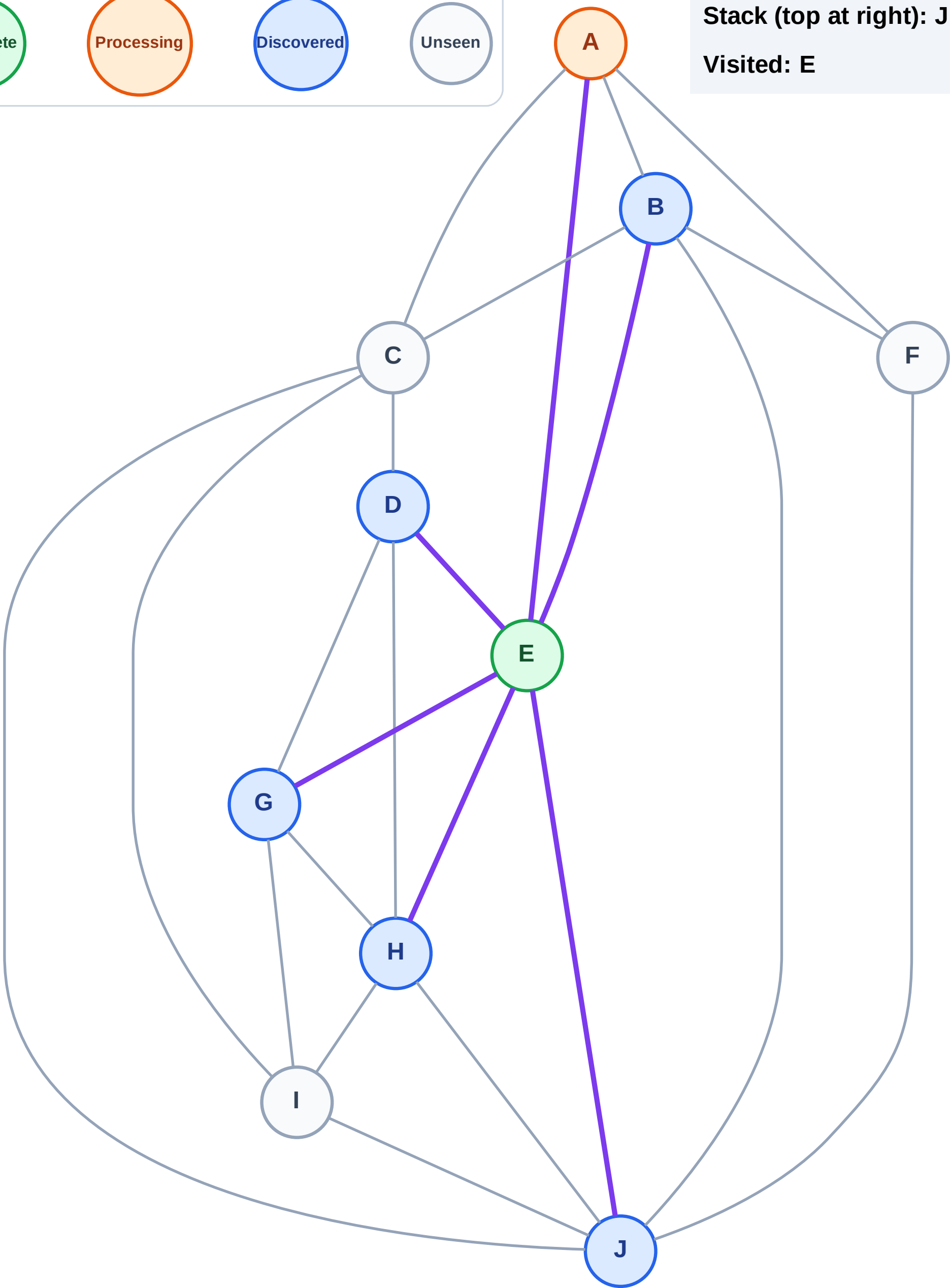
Processing

Discovered

Unseen

Stack (top at right): J | H | G | D | B

Visited: E



DFS Traversal

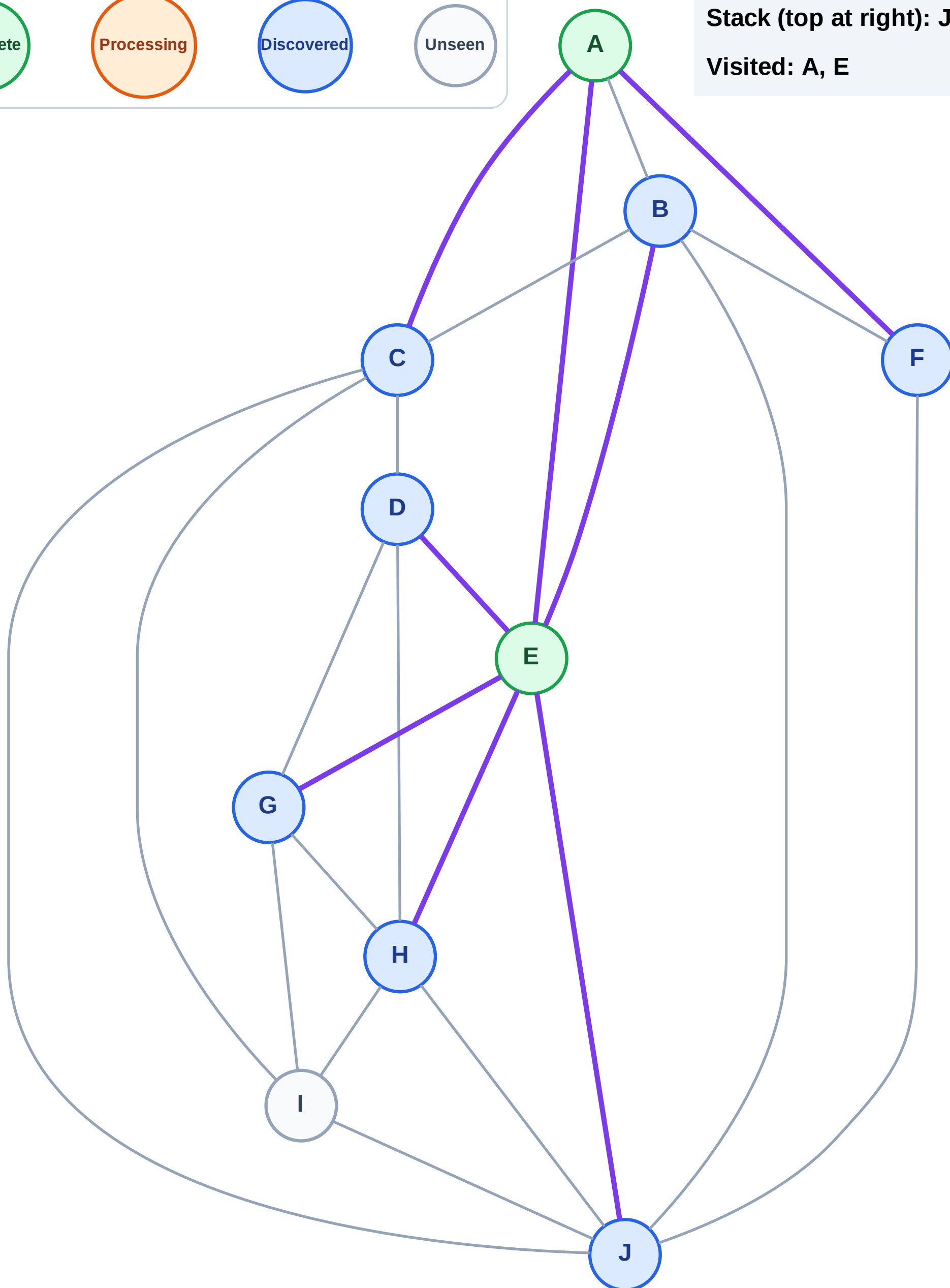
Step 5: Discover F, C from A

Legend



Stack (top at right): J | H | G | D | B | F | C

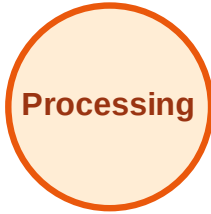
Visited: A, E



DFS Traversal

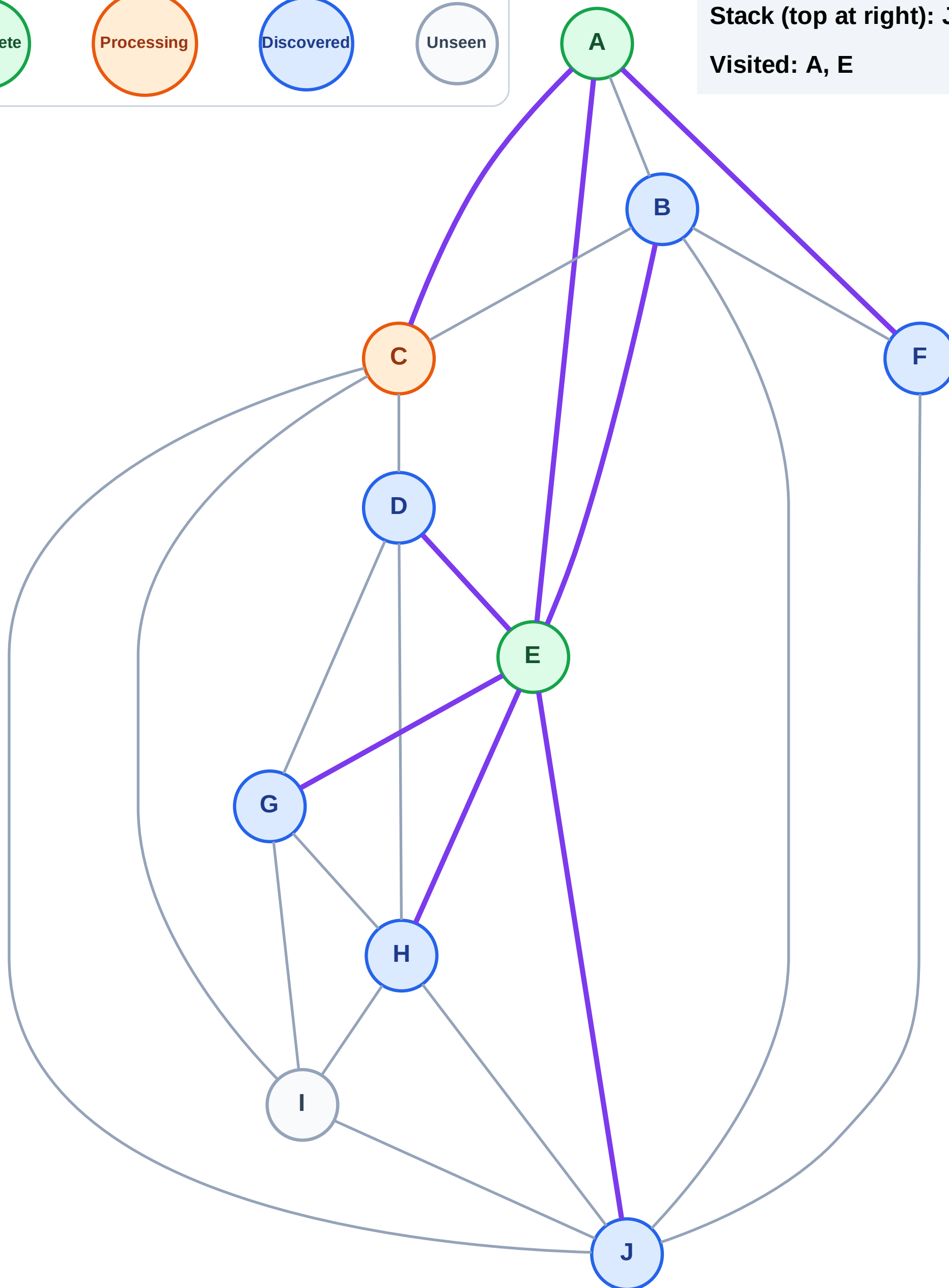
Step 6: Process node C

Legend



Stack (top at right): J | H | G | D | B | F

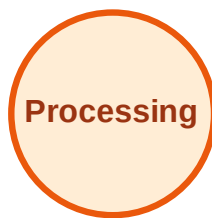
Visited: A, E



DFS Traversal

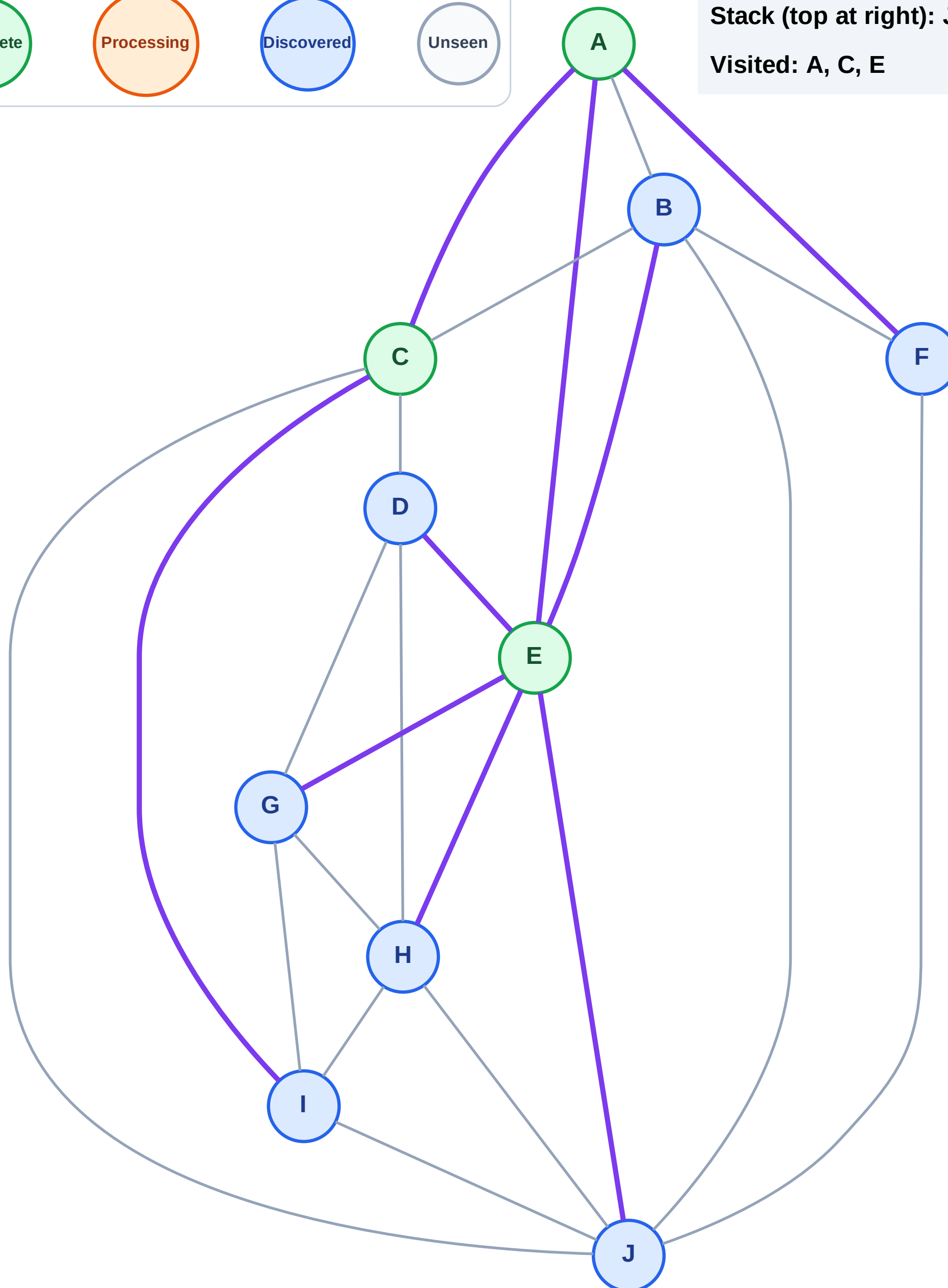
Step 7: Discover I from C

Legend



Stack (top at right): J | H | G | D | B | F | I

Visited: A, C, E



DFS Traversal

Step 8: Process node I

Legend

Complete

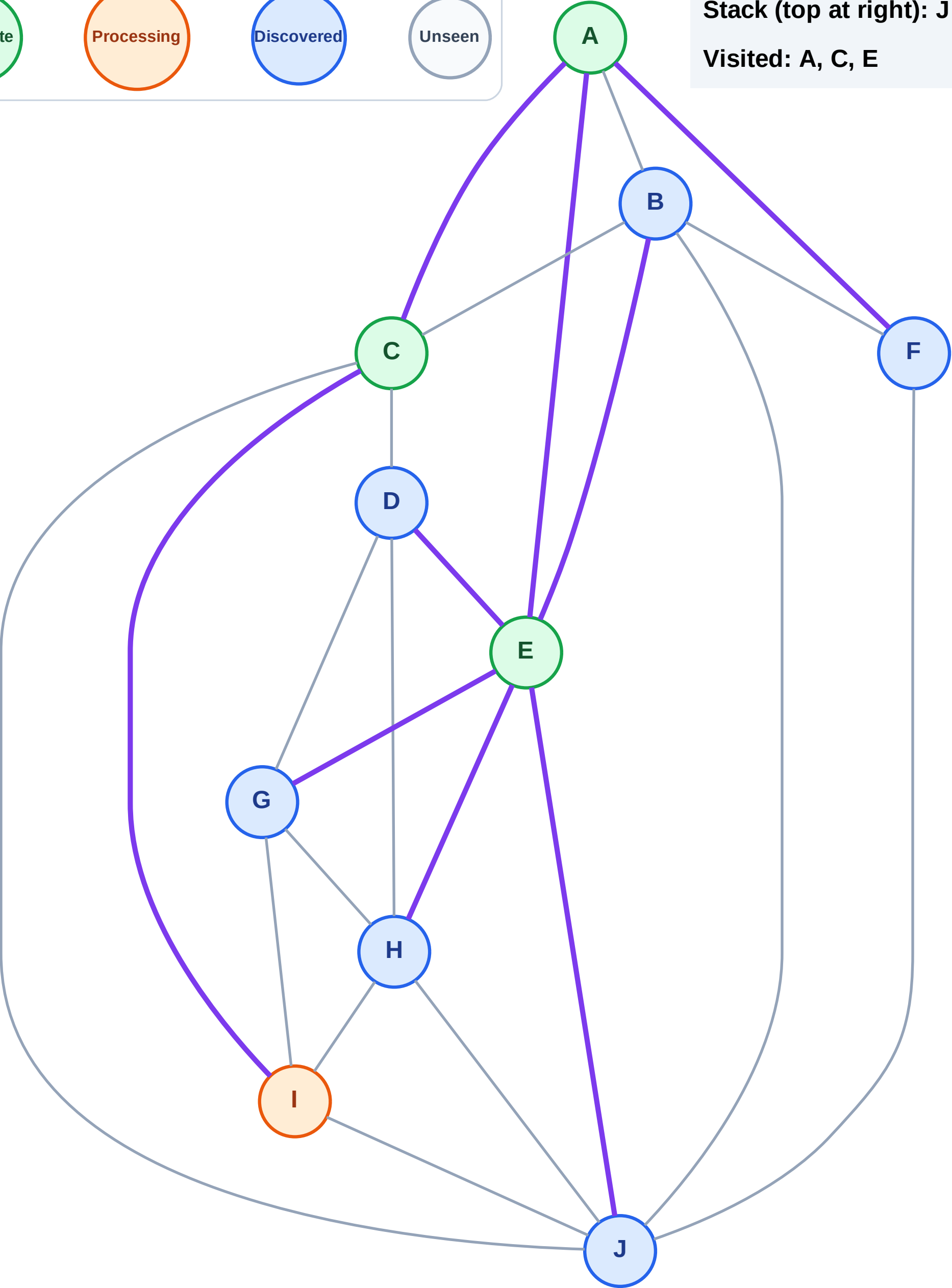
Processing

Discovered

Unseen

Stack (top at right): J | H | G | D | B | F

Visited: A, C, E



DFS Traversal

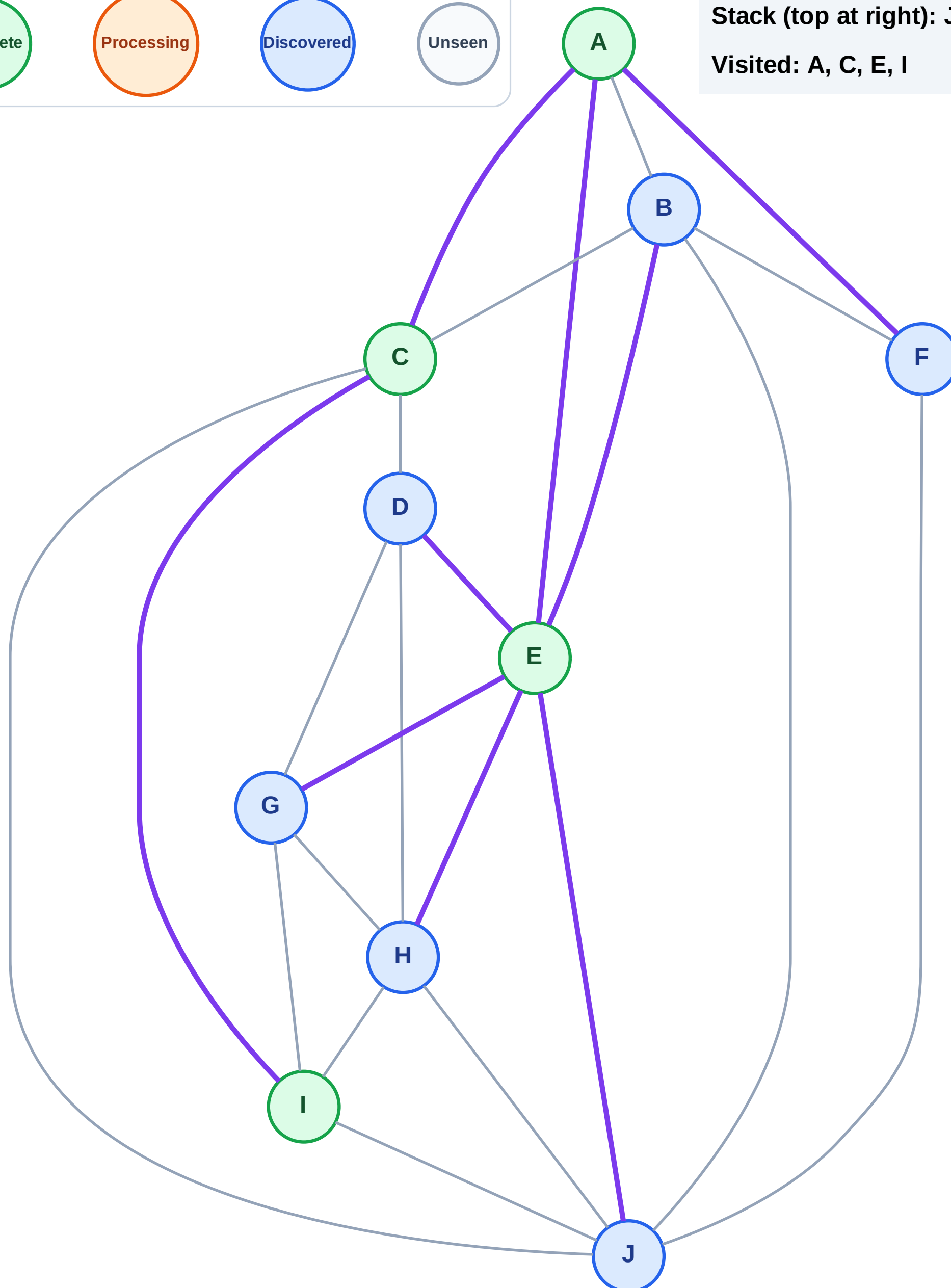
Step 9: No new neighbors from I

Legend



Stack (top at right): J | H | G | D | B | F

Visited: A, C, E, I



DFS Traversal

Step 10: Process node F

Legend

Complete

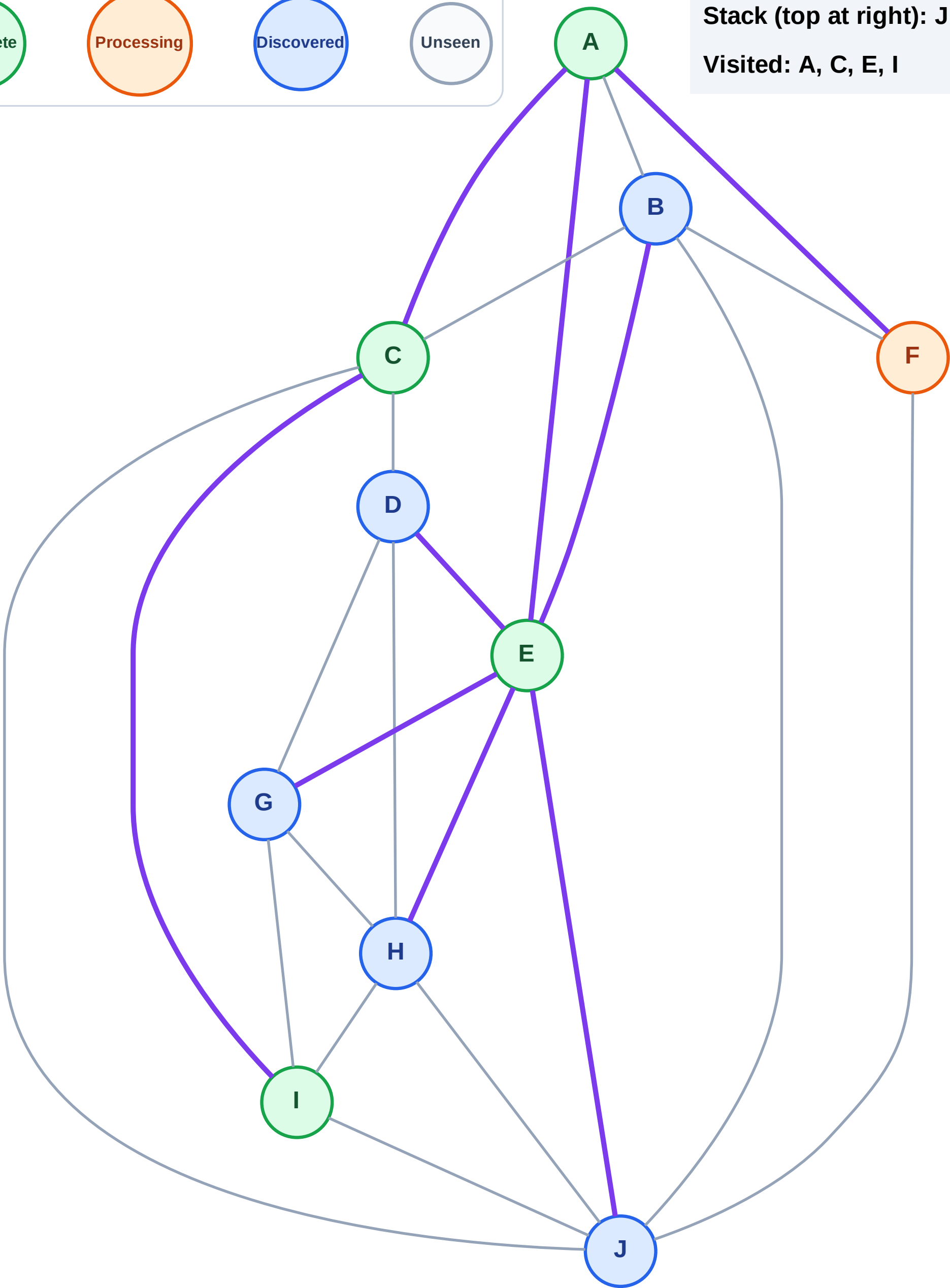
Processing

Discovered

Unseen

Stack (top at right): J | H | G | D | B

Visited: A, C, E, I



DFS Traversal

Step 11: No new neighbors from F

Legend

Complete

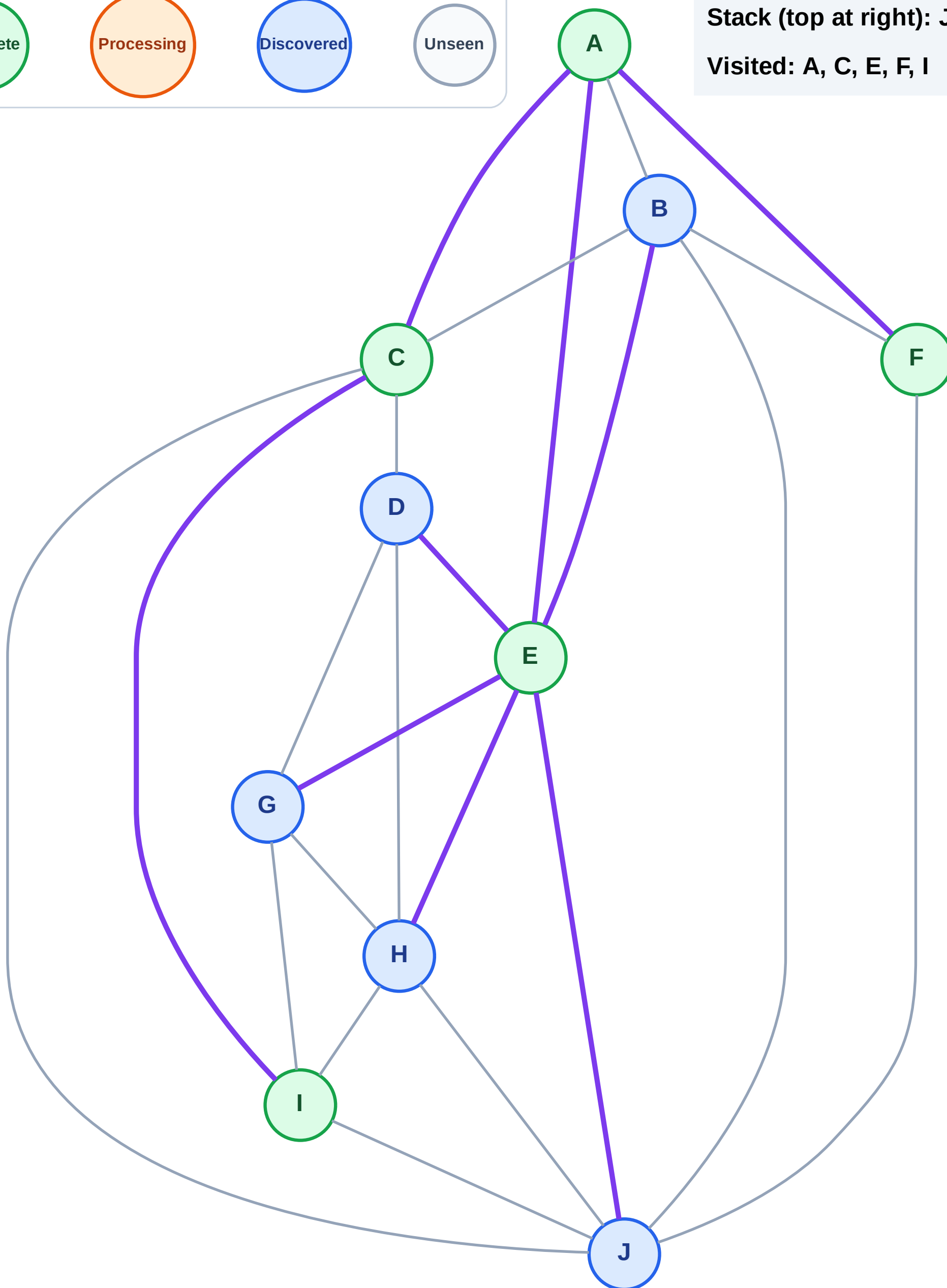
Processing

Discovered

Unseen

Stack (top at right): J | H | G | D | B

Visited: A, C, E, F, I



DFS Traversal

Step 12: Process node B

Legend

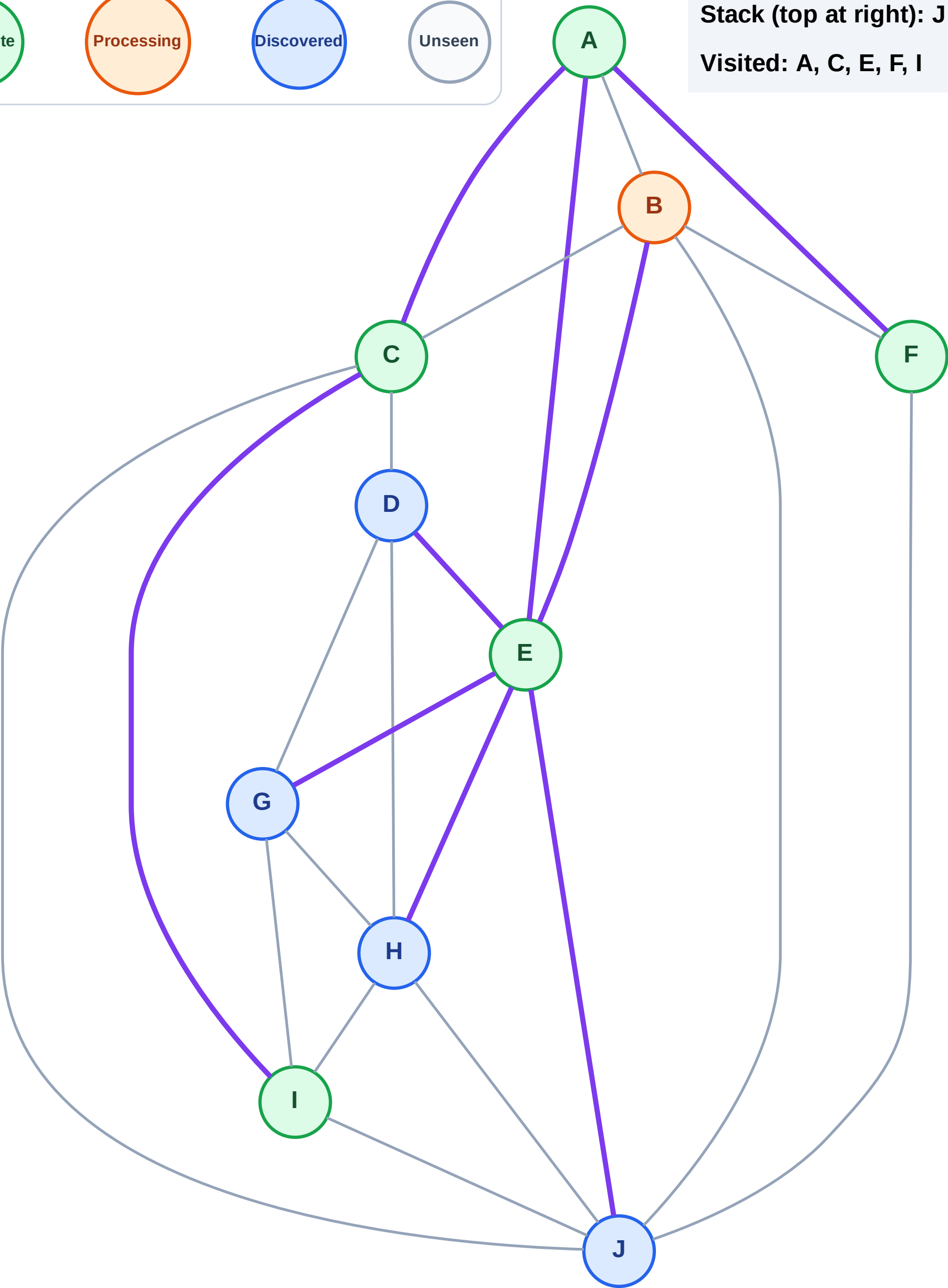
Complete

Processing

Discovered

Unseen

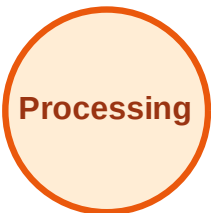
Stack (top at right): J | H | G | D
Visited: A, C, E, F, I



DFS Traversal

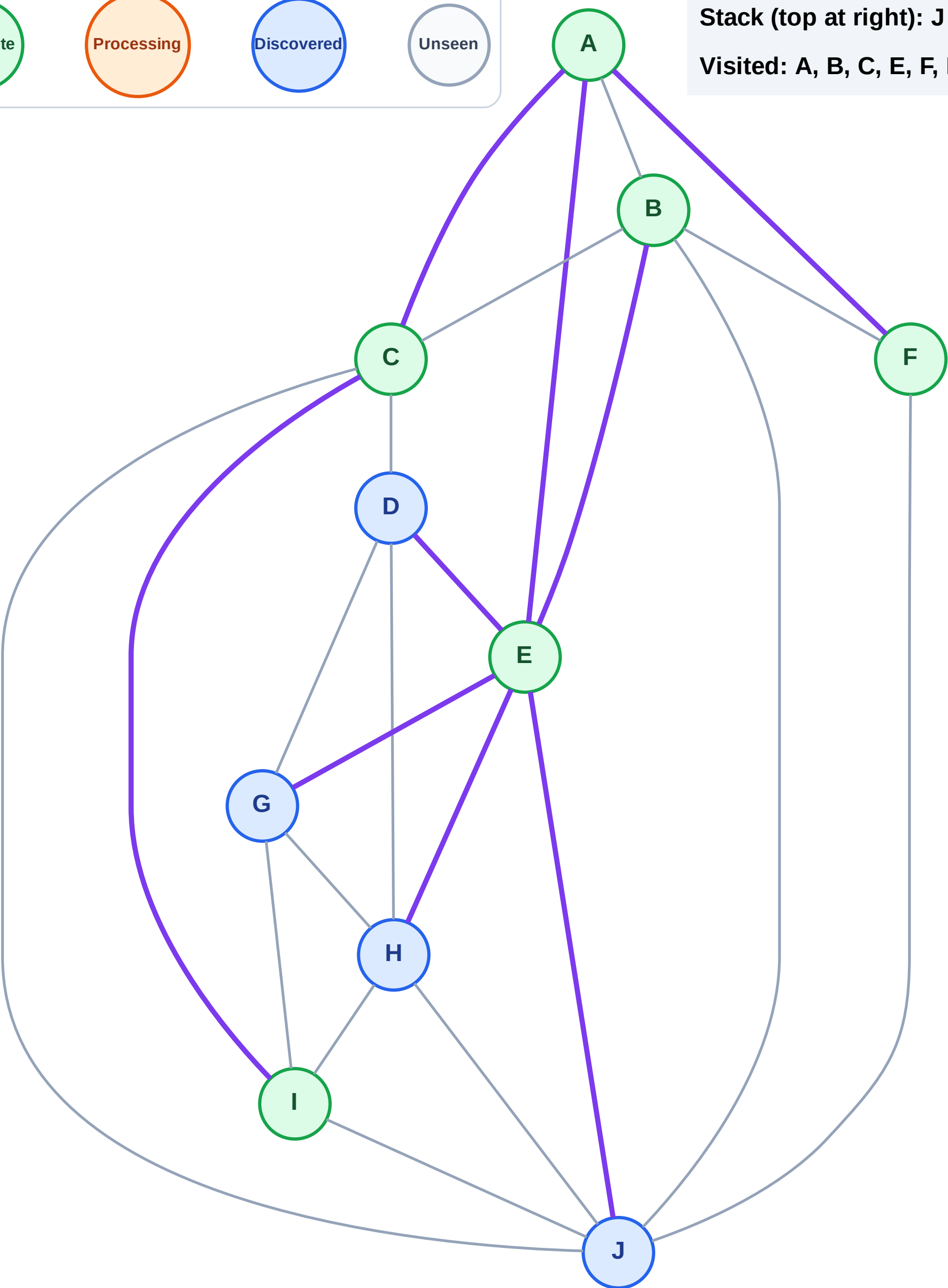
Step 13: No new neighbors from B

Legend



Stack (top at right): J | H | G | D

Visited: A, B, C, E, F, I



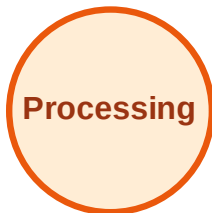
DFS Traversal

Step 14: Process node D

Legend



Complete



Processing



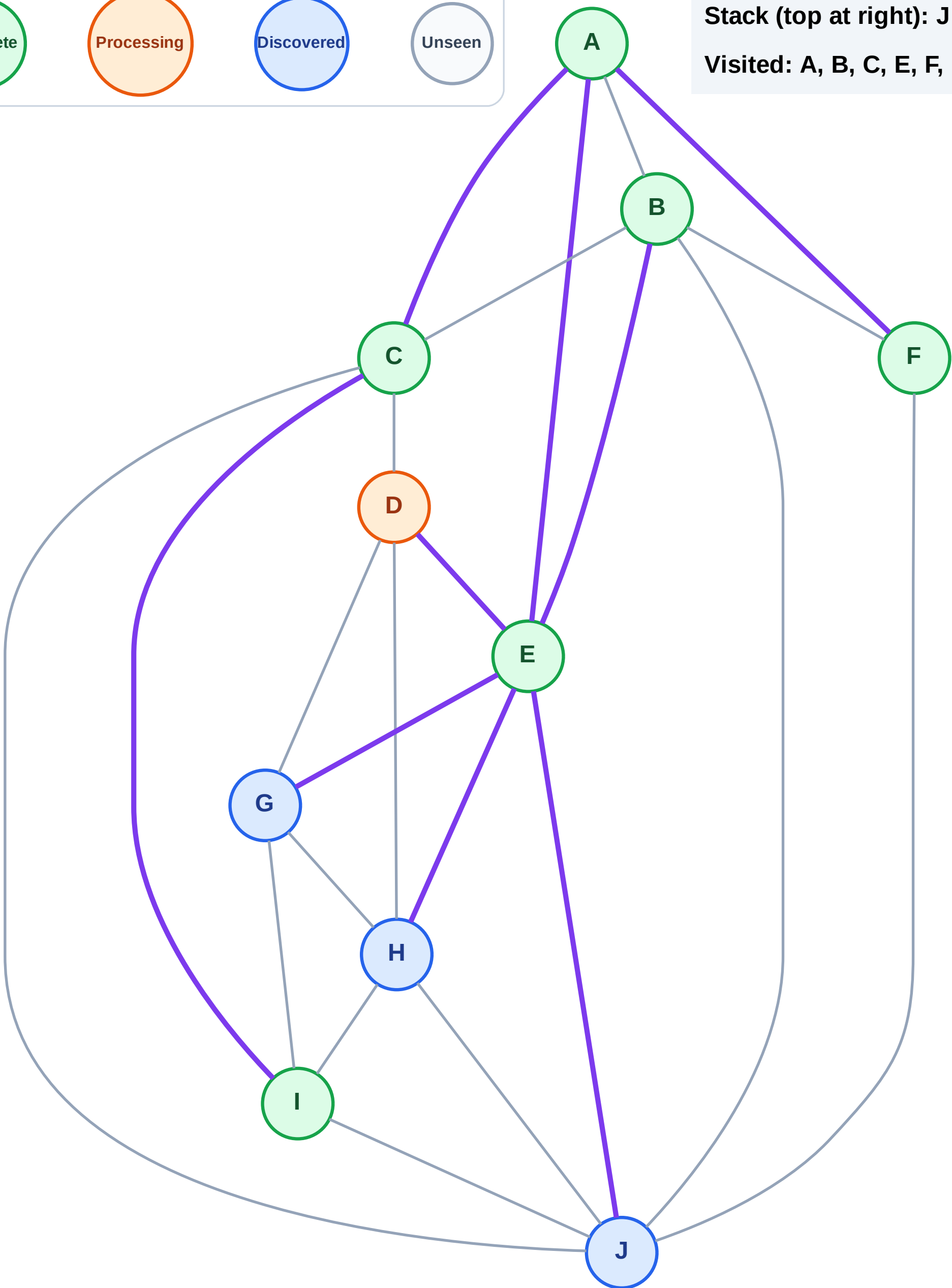
Discovered



Unseen

Stack (top at right): J | H | G

Visited: A, B, C, E, F, I



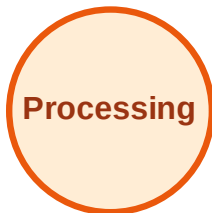
DFS Traversal

Step 15: No new neighbors from D

Legend



Complete



Processing



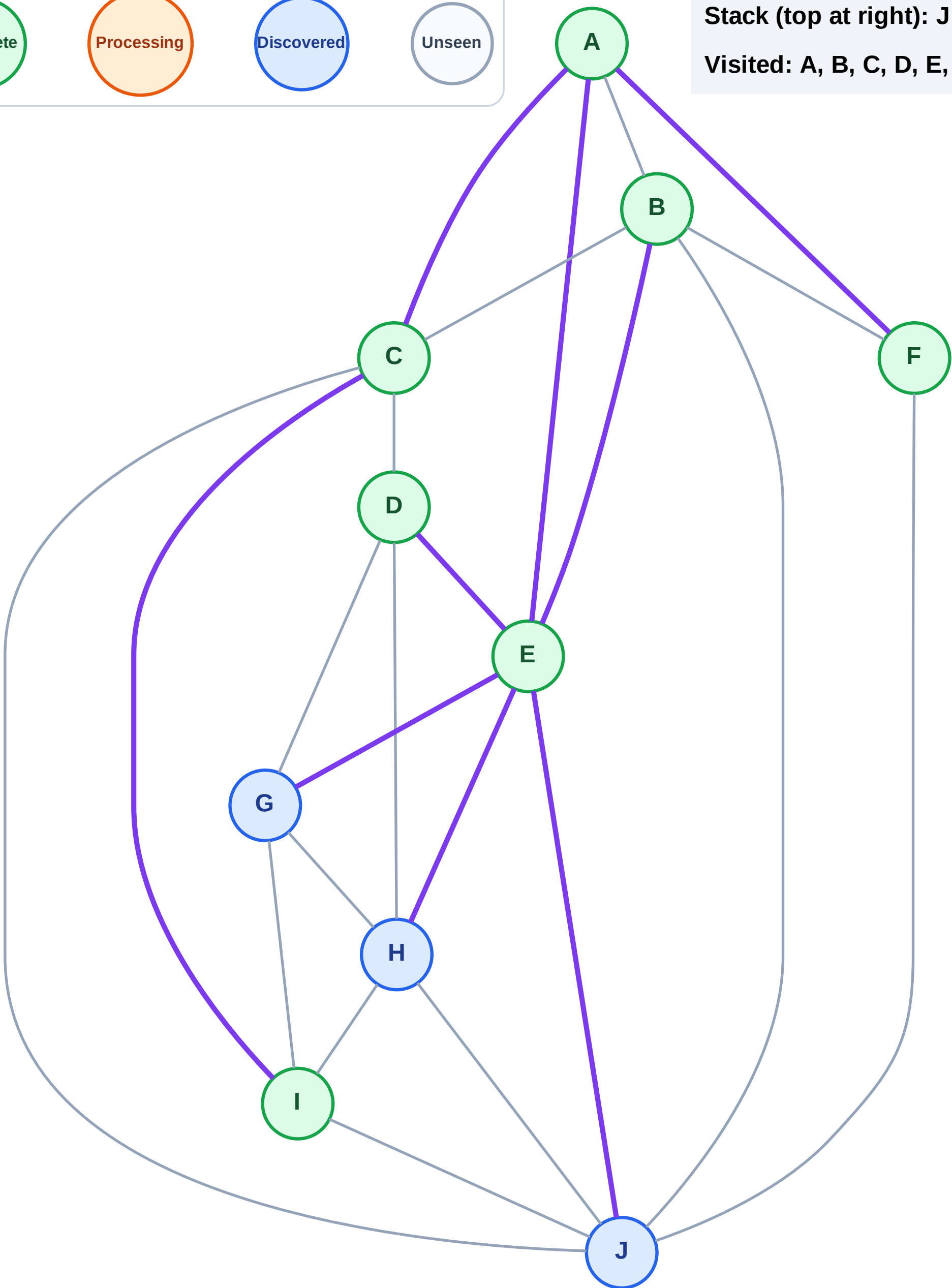
Discovered



Unseen

Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 16: Process node G

Legend

Complete

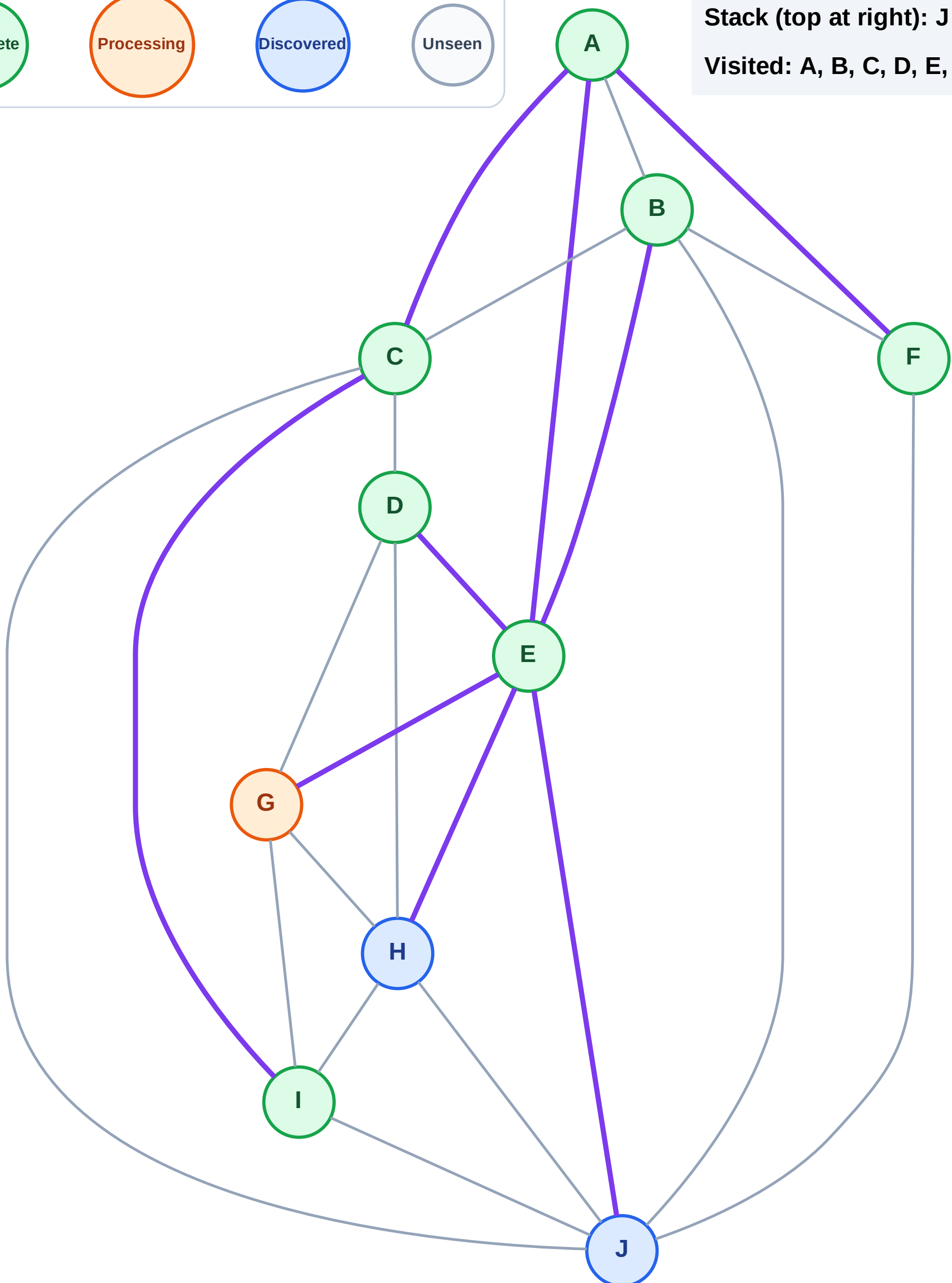
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, I



DFS Traversal

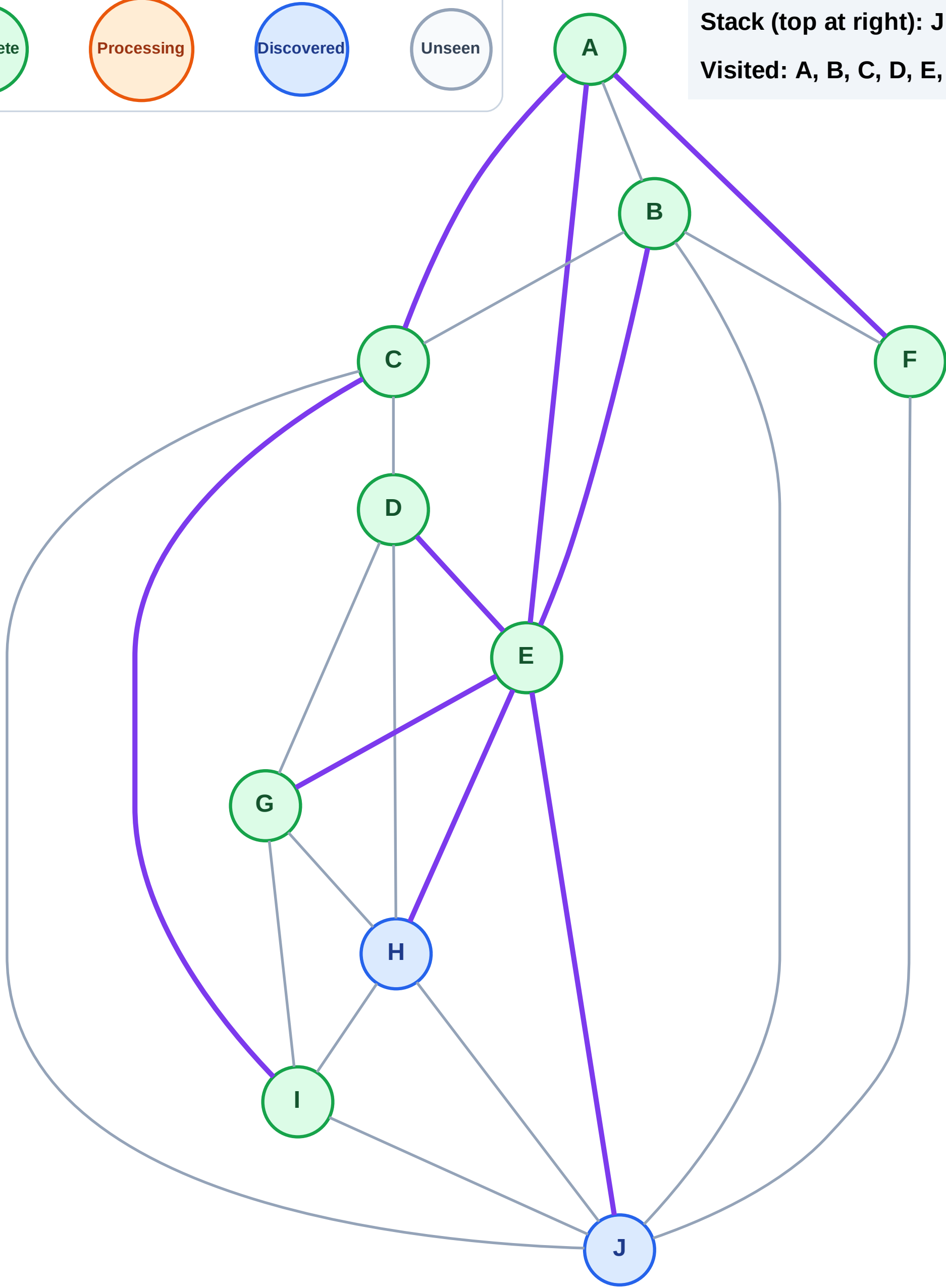
Step 17: No new neighbors from G

Legend



Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



DFS Traversal

Step 18: Process node H

Legend

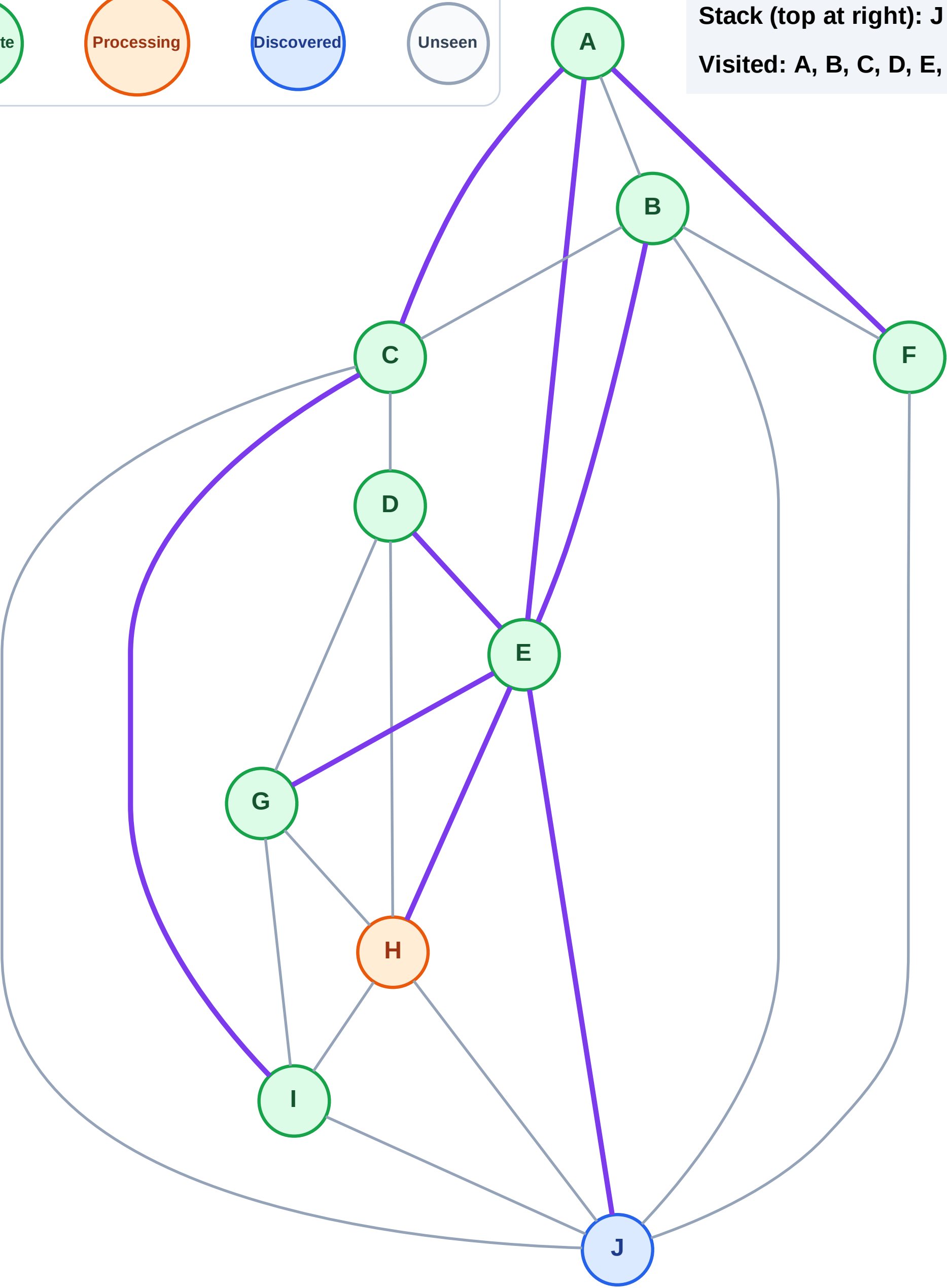
Complete

Processing

Discovered

Unseen

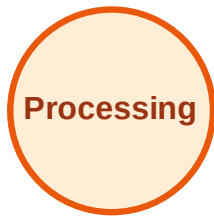
Stack (top at right): J
Visited: A, B, C, D, E, F, G, I



DFS Traversal

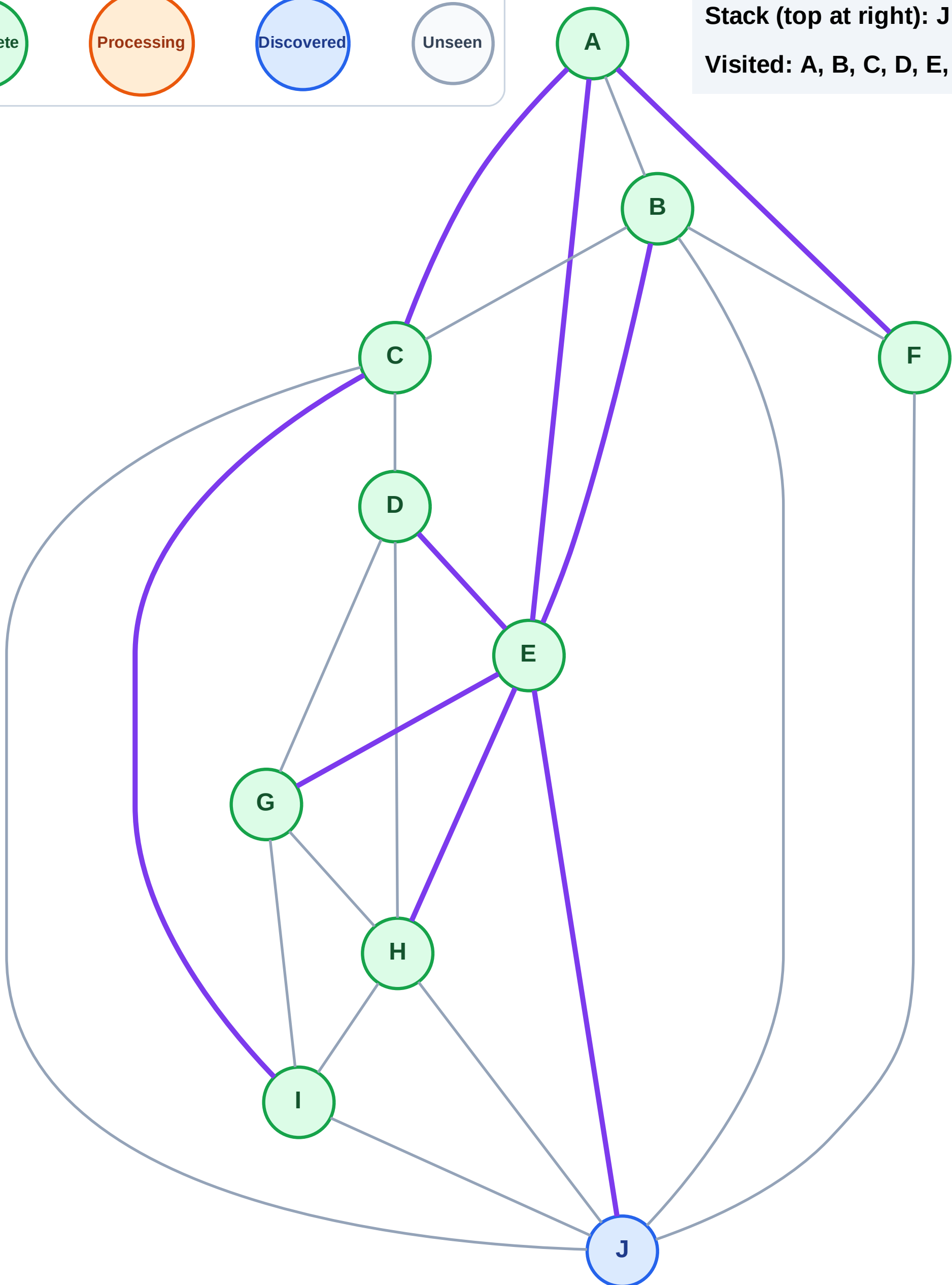
Step 19: No new neighbors from H

Legend



Stack (top at right): J

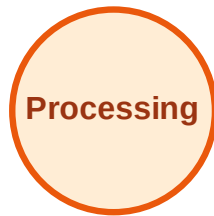
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

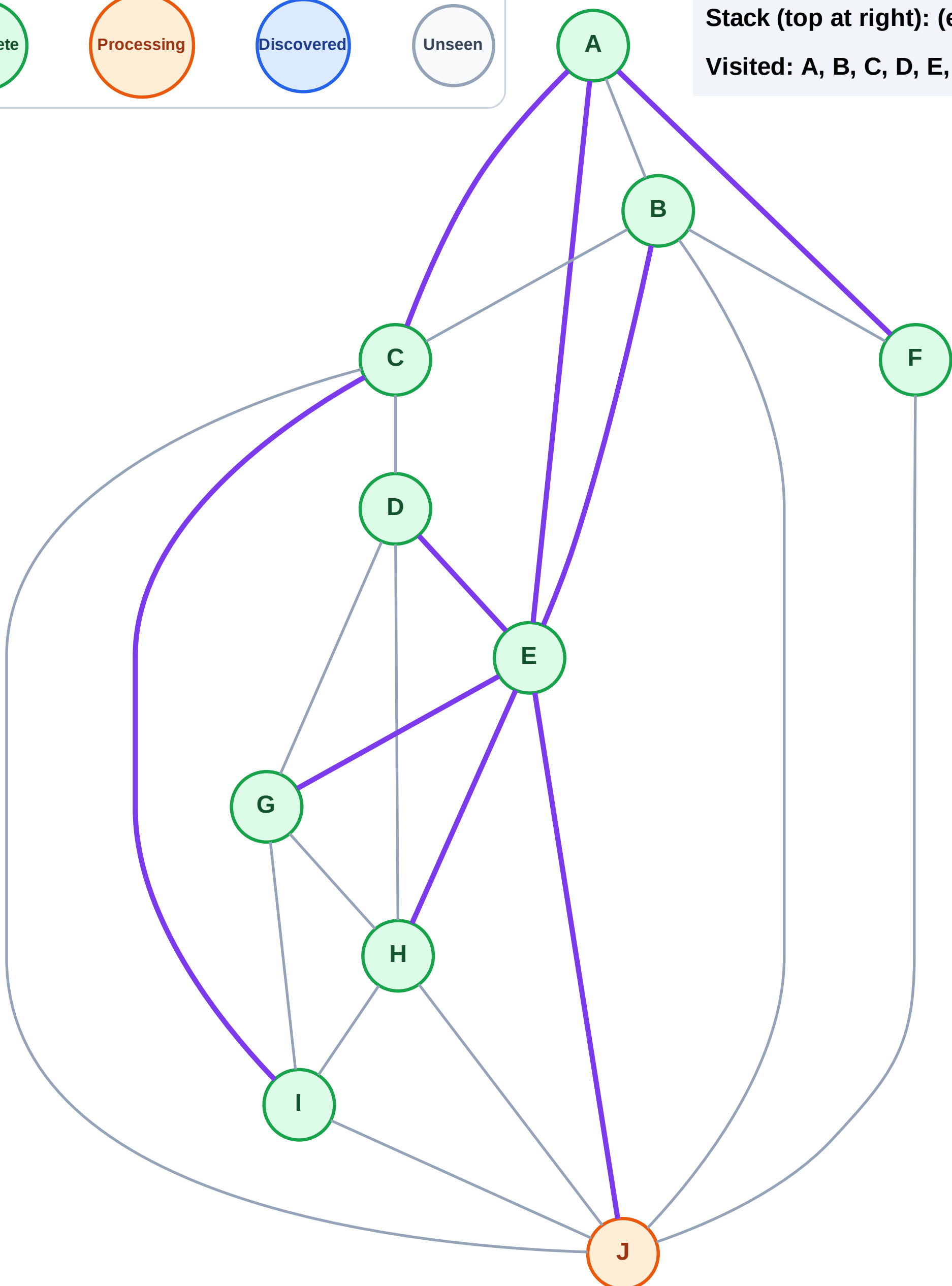
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



Step 21: No new neighbors from J

Step 21: No new neighbors from J

